

CLQT: A Closed-Loop, Cost-Aware, Strategy-Consistent Benchmark for Diagnostic Evaluation of LLM Portfolio-Management Agents

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Abstract

LLM agents are increasingly cast as autonomous portfolio managers, and benchmarks have moved their evaluation from financial question-answering to sequential trading [1, 2, 3]. Most closed-loop trading benchmarks still emphasize a leaderboard of returns over a fixed window, an incomplete proxy because a period’s return is dominated by the market path and apparent agent alpha can largely dissolve once look-ahead bias (information leakage) is controlled, leaving mostly passive factor exposure [8, 9]. Such a ranking certifies neither the soundness of an agent’s reasoning, nor whether it holds a consistent strategy, nor the durability of its edge. We introduce **CLQT**, which reframes closed-loop trading evaluation as *diagnosis* rather than ranking — following the broader turn in agent evaluation toward capability *profiles* over single scores [14, 23, 24] — an instrument that localizes *where* and *why* an agent’s process succeeds or fails.

CLQT is a fully **closed-loop**, **cost-aware**, **strategy-consistent**, temporally-gated environment whose agents run a **five-stage cycle** — gather, synthesize, construct a mandate-aware portfolio, execute under realistic costs, then reflect — in the reason-act-reflect tradition [15, 16]. Each round emits a complete **DecisionRound** sealed into a recompute-verifiable **hash chain**, so every metric is reconstructable from the trail with no added instrumentation. Six pillars form the substrate: a hard **TimeGate** that treats leakage as a failed precondition, institutional **transaction- and financing-cost modeling**, **strategy-consistency scoring** across rounds, a **three-tier consolidating memory**, a **Model-Context-Protocol tool layer** [20], and a **mandate-aware synthesis pipeline**. The same agent runs as a constrained investment committee of specialized roles or a single full-autonomy orchestrator, making **process scaffolding** an experimental variable. From the audit trail we compute a **five-axis capability scorecard (APM-CS)** — Coherence, Acuity, Composure, Discipline, Reliability — each a finance instantiation of a general agent capability, with coherence judged partly by a held-out, out-of-cohort LLM to curb self-preference bias [13]. We validate it on a contamination-controlled multi-model backtest with an ablation grid and a live broker track on unseen, post-cutoff data, disciplining every claim with a repeated-run noise floor. Three findings illustrate what the diagnosis-first read yields: the capability leader is not the Sharpe leader (the nominal Sharpe “winner” is exposed as a 5-of-26-round reliability artifact); mechanical signal-action agreement exceeds a held-out judge’s coherence score by a stable margin on both tracks (+0.30 backtest, +0.23 live) — agents trade in the right *direction* while their allocations fail to follow their own stated analysis;

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and module value registers on the capability axes when returns cannot separate it. CLQT thus separates *outcome* from *capability* across backtest and live; its lasting contribution is not a ranking of today’s models but a durable, extensible map of agent competencies and limitations.

Keywords: large language models, LLM agents, agent evaluation, closed-loop trading, algorithmic trading, portfolio management, backtesting, paper trading, transaction costs, strategy consistency, point-in-time data, look-ahead bias, LLM-as-a-judge, capability diagnostics.

1. Introduction

Recent benchmarks for LLM-driven trading have advanced from static financial QA toward sequential decision-making [1, 2, 3]. Yet two gaps persist. The first is methodological coverage: no existing benchmark simultaneously enforces **point-in-time data semantics**, models realistic **transaction and financing costs** across asset classes, evaluates **portfolio-level** (not per-asset) decisions, measures **strategy consistency** across rounds, accumulates **structured multi-tier memory**, and treats **tool orchestration** as a benchmarkable capability. An agent that achieves high returns by inadvertently accessing future data, ignoring execution costs, or drifting from its mandate provides no credible signal about LLM capability.

The second gap is interpretive. The dominant evaluation idiom in LLM trading is the *leaderboard*: rank agents by Sharpe ratio or cumulative return over a fixed period. Yet a returns-only leaderboard is insufficient for a research benchmark. A single period is dominated by the realized market path and by which risk profile happened to suit it; a model can top the table for reasons unrelated to the quality of its decisions. A credible model ranking would also require many models, regimes, and repeated runs, beyond the scope of one study. We therefore argue for **diagnosis before ranking**: a closed-loop benchmark should first use the decision trail to localize where and why the gather→analyze→decide→execute→reflect process breaks down.

CLQT is built to be that instrument. Its design thesis is that a model should be called “superior” only when it dominates the relevant capability axes consistently across sub-periods. The benchmark therefore supplies a reusable map of limitations rather than treating one outcome as decisive. We compute that map as a **five-axis diagnostic capability scorecard** from the **audit trail** (§6.4). A **recompute-verifiable hash chain** makes the provided trail independently integrity-checkable under the threat model in §4.4. Figure 1 summarizes the design.

The contributions of this paper are:

1. A **diagnosis-before-ranking reframing** of closed-loop trading evaluation: the contribution is an instrument that maps agent limitations, with an explicit standard (“consistent dominance across axes and sub-periods”) that resists premature ranking from a single outcome metric.
2. A **five-axis capability scorecard (APM-CS)** — Coherence, Acuity, Composure, Discipline, Reliability — each a finance instantiation of a general agent-evaluation capability, synthesized from existing audit fields with no new instrumentation.

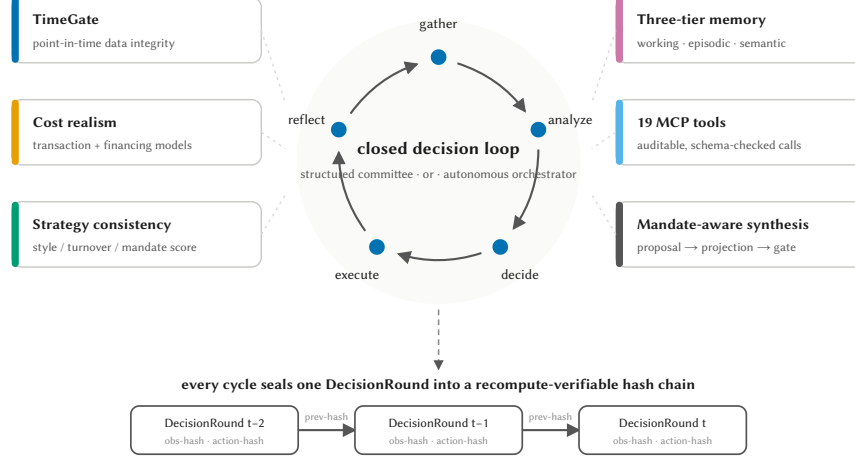


Figure 1. *CLQT graphical abstract — six design pillars, from temporal integrity to three-tier memory, surround the five-stage closed decision loop; every cycle seals one DecisionRound into a hash-linked audit chain.*

3. A **held-out LLM-as-a-judge** for the Coherence axis (non-cohort, reasoning-enabled, cached), guarding against self-preference bias — with the headline coherence gap corroborated by a second, different-family held-out judge (Spearman $\rho = 0.65$), so the finding is not an artifact of one judge.
4. A **recompute-verifiable audit hash chain**: each round’s observation and action hashes are recomputed on verification and linked to the prior round’s action hash, exposing inconsistent modification of the provided store under the threat model in §4.4.
5. A **closed-loop benchmark substrate** with strict temporal integrity, explicit multi-component cost modeling, strategy-consistency measurement, three-tier memory, 19 MCP tools, mandate-aware synthesis, 3-D scouting, intertemporal sentiment trajectories, and verified live broker execution (carried forward and consolidated from prior versions).
6. A **grounded demonstration**: a chain-verified year-long bi-weekly campaign (5 models $\times$ 2 skill modes, repeated runs for the reliable models) plus a 13-configuration ablation grid (the full-module baseline, eight single-module / cost-tier variants, and four multi-module cluster knockouts, repeated-run), a held-out Coherence judge, and a live broker track — showing the instrument separates outcome from capability, surfaces model-specific limitations, and explicitly bounds what the evidence can claim.

Roadmap. The paper is organized so that each layer is in place before it is used. §2 locates CLQT in the benchmark landscape and names the agent-infrastructure and cognitive foundations its design borrows. §3–§5 describe the substrate that *produces* the

evidence: the closed-loop system (§3), the temporal-integrity and audit machinery that make its numbers trustworthy (§4), and the backtest and live protocols that generate them (§5). §6 defines what is measured from that substrate — the five-axis scorecard (§6.4, D1–D5), its held-out judge protocol (§6.4.1), and the behavioral method (§6.5). §7 then presents the evidence in the order the concepts were built: the year-long backtest campaign (§7.3–§7.6), the ablation grid (§7.7), and the live validation and behavioral findings (§7.10–§7.11), with the research questions and their answers mapped up front in §7.1. §8 discusses what the findings mean beyond finance, and §8.7 bounds what they may claim.

2. Related Work and Positioning

2.1 Benchmark Landscape

We survey seven representative closed-loop trading benchmarks and map them against CLQT on seven *core* features. To keep the comparison conservative, the map credits any partially-supported feature as fully supported for prior work, and records only feature *presence* — not quality or depth.

Table 1. *Core-feature coverage across closed-loop LLM trading benchmarks, ordered by year. Features: PiT = point-in-time data enforcement; Cost = transaction-cost modeling; Port = portfolio-level (multi-holding) evaluation; Multi = multi-asset; Mem = cross-round memory; Tools = tool-orchestration evaluation; Live = live-track support. ✓ = supported, — = absent; a partially-supported feature is credited as ✓ for prior benchmarks. Covered is a pure feature-credit tally (of 7), not a quality score. CLQT’s further contributions — strategy-consistency measurement, a verifiable audit trail, and an ablation framework — are CLQT-specific and omitted from this shared-feature view.*

Benchmark	Year	PiT	Cost	Port	Multi	Mem	Tools	Live	Covered
InvestorBench [5]	2024	—	—	—	✓	✓	—	—	2/7
StockBench [1]	2025	✓	—	—	—	—	—	—	1/7
FINSABER [6]	2025	✓	✓	—	—	—	—	—	2/7
DeepFund [4]	2025	✓	—	✓	—	—	—	✓	3/7
AI-Trader [3]	2025	✓	—	—	✓	—	✓	✓	4/7
LiveTradeBench [2]	2025	✓	—	✓	✓	—	—	✓	4/7
PortBench [7]	2026	✓	—	✓	✓	—	—	—	3/7
CLQT (ours)	2026	✓	✓	✓	✓	✓	✓	✓	7/7

These benchmarks target different universes, periods, metrics, and goals, so a mark records the *presence* of a feature, not performance — and each prior benchmark is strong on the facet it was built for: leakage control (StockBench, DeepFund), live realism (LiveTradeBench, AI-Trader), multi-asset pipeline scoring (PortBench, LiveTradeBench). Several implement their target feature more thoroughly than CLQT does. CLQT’s claim is only breadth of coverage in one closed loop, which the matrix shows; it is not that CLQT is uniformly better than any of them.

Recent work (2025–2026), read closely. Four strands of very recent work each sharpen one facet of the problem; reading what each *does and does not* measure is the clearest way to locate CLQT.

Full-pipeline scoring. PortBench [7] is the closest in spirit — it evaluates the entire portfolio-management *pipeline* with correlation-aware scoring and a cross-stage error-propagation metric. But it scores that pipeline as a backtest over historical windows, stage-by-stage, rather than as a *closed loop* — without the persistent memory and tool use a live agent runs on. Its central move — decomposing *where in the pipeline* error originates rather than reporting a single score — is precisely the diagnostic stance CLQT adopts; CLQT extends it to a *closed loop* with memory, MCP tools, realistic costs and a live track (§3, §5), and replaces stage-accuracy with a five-axis capability profile.

Returns-centric autonomous trading. StockBench [1] and AI-Trader [3] run autonomous stock-trading agents end-to-end and report returns-centric outcomes (cumulative return, Sharpe, drawdown) rather than a decision-*process* capability profile. Their demonstration that capable models can trade profitably motivates CLQT’s complementary question — *is the profit coherent with the agent’s stated reasoning?* — which §7 shows it frequently is not (the Sharpe leader is a 5-of-26-round artifact).

Integrity and leakage. A focused line documents the failure mode CLQT’s TimeGate exists to prevent: Profit Mirage [8] shows that apparent LLM-agent alpha largely collapses once information leakage is removed, and memory-controlled masking [9] reports that returns under leakage-controlled evaluation are substantially explained by passive market and style exposure rather than stock-selection alpha. These results motivate CLQT’s hard temporal gate (§4) and per-run integrity ledger: we treat point-in-time enforcement as a *precondition for a trustworthy number*, not an optional safeguard.

Memory as a measured variable. Memory-controlled trading benchmarks [9] (“from knowing to doing”) isolate the variable CLQT promotes to a first-class axis — whether an agent can convert recalled experience into improved action — which CLQT measures directly through its three-tier memory (§3.6) and the memory ablation (§7.7).

Synthesis. No existing benchmark combines these facets. PortBench has the pipeline view but is backtest-only and not a closed loop; the returns benchmarks have autonomy and sometimes a live track but report a scalar; the integrity line fixes one precondition in isolation; the memory line isolates one variable. CLQT occupies the position none of them fills: a closed-loop, cost-aware, point-in-time, memory- and tool-bearing system that scores the decision *process* as a five-axis capability profile across both backtest and live (Table 1).

2.2 Key Gaps Addressed

No cost-awareness. Prior benchmarks either ignore transaction costs [1, 2, 3, 4] or apply a single flat commission [6]. CLQT implements four calibrated cost models (§3.4) across four asset classes and four cost tiers, plus daily financing-cost accrual.

No cross-round learning evaluation. InvestorBench [5] mentions memory but does not formalize or measure it. CLQT’s three-tier memory (§3.6) stores structured event-action-outcome records that decay and consolidate.

No tool synthesis evaluation. AI-Trader [3] mandates tool-only execution but does not evaluate orchestration quality. CLQT logs every MCP tool call with hashes and

latency.

No intertemporal information accumulation. All prior benchmarks treat news as single-round context. CLQT accumulates news across cycles in a persistent store (§3.8) and measures whether agents reason about sentiment *dynamics*.

No multi-dimensional candidate scouting. Prior work screens by price momentum alone, if at all. CLQT’s 3-D SCOUT stage (§3.3) adds earnings surprise and macro-correlation.

No bridge-to-live execution. Most benchmarks stop at simulated fills. CLQT implements a complete live broker integration (§5.2) with paper-trading verification.

2.3 The LLM-Agent Infrastructure CLQT Builds On

Before it is a finance benchmark, CLQT is an LLM-agent system, and almost none of its machinery originates in financial AI — the loop, the memory, the tools, and the multi-agent committee are all adapted from the general agent-infrastructure literature of 2022–2025. We name that provenance explicitly, because it makes the design legible: each component exists to instantiate a mechanism that line of work showed to matter, and CLQT’s contribution is to make that mechanism *measurable* in a cost-bearing, temporally-gated decision process.

Reasoning-acting loop. The five-stage loop follows the reasoning-and-acting paradigm of ReAct [15] and the self-critique loop of Reflexion [16]. CoALA (Cognitive Architectures for Language Agents) [19] supplies the unifying frame — language agents as production systems with explicit memory, action, and decision modules — that CLQT’s stage decomposition mirrors almost one-to-one (and connects directly to the ACT* grounding of §2.4).

Memory — borrowed from general agent infrastructure, not finance. The intuition for CLQT’s three-tier memory comes from *outside* trading entirely. Generative Agents [17] showed that a *memory stream* with periodic *reflection* and relevance-weighted *retrieval* yields coherent long-horizon behavior in a sandbox of simulated humans; MemGPT [18] reframed context management as operating-system-style paging between a working context and external stores; and CoALA [19] organizes agent memory into working, episodic, semantic, and procedural tiers. CLQT adapts exactly this lineage — a memory stream that consolidates event→action→outcome records, decays, and is retrieved by relevance — to a portfolio ledger, with extracted “lessons” as reusable procedural memory. The novelty is not the mechanism but treating “*does recalled experience improve the next decision?*” as a measured axis rather than an assumed benefit (memory ablation, §7.7) — answering for trading the question memory-controlled benchmarks [9] pose in general.

Tool use. That agents should *act through tools* rather than emit answers directly is the lesson of the tool-augmented-LLM line, exemplified by ToolLLM [21]; CLQT operationalizes it with a Model-Context-Protocol [20] tool layer (19 quantitative tools) — every tool call hashed, timed, and auditable, so tool *use* becomes evidence rather than a black box.

Multi-agent committee and the structured scaffold. The role-decomposed committee (PM,

Risk Officer, ...) draws on the general multi-agent-framework literature; the structured skill mode is inspired most directly by MetaGPT [22], whose central finding is that encoding standard operating procedures into the agent workflow reduces cascading error relative to free-form coordination. That result is the *intuition behind CLQT’s structured-vs-autonomous contrast* (§7.6): we do not assume scaffolding helps — we make the SOP-vs-autonomy trade-off an experimental variable and measure which models it helps, which it constrains, and on which axes. Within finance, the committee also builds on multi-agent trading frameworks like TradingAgents [10] and memory-augmented trading agents like FinMem [11].

Evaluation as a capability profile. The decision to report a *profile* rather than a scalar is itself borrowed: HELM [14] argued for holistic, multi-metric evaluation over a single leaderboard number, and agent benchmarks such as AgentBench [23] and τ -bench [24] score agents across tasks and *trajectories* rather than final answers. CLQT is the finance instantiation of that turn — five axes, a hash-chained audit trail, and a held-out judge — specialized to a closed-loop, cost-bearing portfolio process.

Judge design and statistical hygiene. Two further practices come from the same evaluation literature. Recent work on LLM-as-a-judge [13] cautions about self-preference bias and the need for non-cohort judges and rubric grounding; CLQT’s Coherence judge is accordingly held out of the evaluated cohort, cached, and reported *alongside* the mechanical signal-action agreement it systematically diverges from (the +0.30/+0.23 per-track gap of §7.10). That divergence — essentially the same on backtest and live — is itself the diagnostic, in the spirit of procedure-aware evaluation [12], which scores the *correctness of the process* rather than only task success. And the instrument is explicit about statistical power: repeated runs with a noise floor read against every ablation Δ , and reliability (format adherence, completion, truncation) as a first-class axis rather than an auxiliary note. CLQT is complementary to broad financial-LLM benchmark suites in treating the *decision process itself* as a measurable capability profile.

2.4 Theoretical and Cognitive Foundations

CLQT’s architecture is not ad hoc: each pillar instantiates an established model of decision-making or cognition, and several of the paper’s findings are most naturally read through these lenses. We make the mappings explicit because they are what lets a *trading* benchmark speak to *general* agent behavior.

The closed loop $\leftarrow$ *decision-cycle theory.* The gather $\rightarrow$ analyze $\rightarrow$ decide $\rightarrow$ execute $\rightarrow$ reflect loop is a finance instantiation of Boyd’s OODA cycle (Observe–Orient–Decide–Act) [26] and of the classic three-level situation-awareness model — *perception* of task-relevant elements, *comprehension* of their meaning, and *projection* of their near-future implications. The mapping is direct: Intelligence Acquisition is perception, Signal Synthesis is comprehension, Portfolio Construction is projection-and-decision. This grounding also *defines* the Acuity axis (§6.4): a Level-1 (perception) failure — attending to salient-but-irrelevant cues — is exactly what Acuity measures when it contrasts an agent’s response to informative signals against a volatility *distractor*.

Reasoning vs. acting $\leftarrow$ *Anderson’s declarative/procedural distinction.* Anderson’s ACT*

theory of cognition [25] separates *declarative* knowledge (factual propositions in semantic memory) from *procedural* knowledge (condition-action production rules compiled only through repeated practice). Anderson’s central claim — that procedural skill is acquired by *compiling declarative precursors through application*, not by declarative retrieval alone — predicts a specific deficit in text-trained models: they encode the *propositions* of a strategy (declarative) but not the *production rules* that let those propositions govern an action (procedural), because the compilation step requires active practice a text corpus cannot supply. This is the theoretical basis for CLQT’s central Coherence axis and, we argue in §8, the most parsimonious explanation of its headline finding — the systematic gap between what an agent *states* and what it *does*.

Reflect-and-learn $\leftarrow$ *experiential-learning theory*. The Attribution & Lesson Extraction stage and three-tier memory operationalize an experiential-learning cycle (concrete experience $\rightarrow$ reflective observation $\rightarrow$ abstract conceptualization $\rightarrow$ active experimentation); their *systems* lineage — the ReAct/Reflexion loop, the MemGPT/CoALA memory tiers — is the one already traced in §2.3. What this subsection adds is the cognitive reading: reflection is, in Anderson’s terms, the benchmark’s attempt to *supply the missing practice*, and whether it helps is an empirical question the memory ablation (§7.7) is built to answer.

Composure $\leftarrow$ *behavioral finance*. The Composure axis measures overreaction to volatility — the LLM-agent analogue of the classic overreaction hypothesis from behavioral finance and of noise-driven trading: a disciplined agent should not churn its book in response to transient volatility. Reading agent churn as overreaction connects CLQT’s per-round turnover-vs-volatility regression to a long empirical tradition.

Synthesis. Grounding the five axes in cognition and decision theory is what makes them portable — the finance task is the *instrument*, but the capabilities it measures are not finance-specific, which is why a portfolio-management benchmark can function as a general agent diagnostic. We map each axis to its construct in §8.2.

3. System Design

CLQT’s design goal is that every mechanism an agent runs on is also a mechanism the evaluation can *measure*. This section walks the substrate in the order a decision round executes it — the loop and its agents (§3.1–§3.2), information gathering and candidate scouting (§3.3), cost-aware execution (§3.4), strategy-consistency measurement (§3.5), memory (§3.6), the decision pipeline (§3.7), and the sentiment trajectory (§3.8) — noting, for each component, which audit fields it emits and which axis of the §6.4 scorecard later reads them.

3.1 Architecture Overview

CLQT operates as a **five-stage closed loop** executed by **TradingEngine**, which orchestrates **Protocol modules** through a **six-agent architecture**. The loop is the OODA / situation-awareness decision cycle of §2.4 made executable — perception (IA), comprehension (MSS), projection-and-decision (MAPC), action (COE), and reflection (ALE) — and the role decomposition follows the multi-agent financial-trading line [10]

while the reflect-and-learn stage instantiates the ReAct [15] / Reflexion [16] paradigm. Each stage is a Python **Protocol interface** — implementations can be swapped without inheritance. All quantitative analysis is performed through **19 typed tools** defined to the **Model-Context-Protocol (MCP)** standard and grouped into four namespaces (market-data, quant-tools, news-sentiment, execution); within the benchmark these run *in-process* against the underlying quant and data libraries — we adopt the MCP tool-definition interface and schemas, not a networked server transport — so every analytical step is an **auditable, schema-checked tool call** rather than free-form text generation (full inventory in Appendix B, Table B1). The five stages are:

1. **Intelligence Acquisition (IA)** — gather market, news, macro, and asset-class-specific data; scout universe candidates.
2. **Multi-Source Signal Synthesis (MSS)** — produce signals, risk assessments, and factor decompositions.
3. **Mandate-Aware Portfolio Construction (MAPC)** — propose and validate target allocations against the investment mandate.
4. **Cost-Optimal Order Execution (COE)** — generate and execute orders with cost-minimizing strategies.
5. **Attribution & Lesson Extraction (ALE)** — attribute performance, extract structured lessons, update memory.

The atomic unit of evaluation is the **DecisionRound**:

```
DecisionRound:
  round_id, as_of_ts, run_id, model_id
  |- IA:      market_snapshot, news_items, sentiment_trajectory, macro_indicators,
  |          asset_class_data, candidate_signals
  |- MSS:     signals, risk_report_pre, factor_attribution
  |- MAPC:    portfolio_before, target_allocation, mapc_delta_proposed/projected,
  |          mapc_adjustments_clipped, universe_changes, target_adherence
  |- COE:     orders, execution_report, portfolio_after, financing_cost
  |- ALE:     risk_report_post, reflection_notes, lessons_generated
  |- Memory:  consistency_score, realized_pnl_this_round, memory_provenance
  |- Cost/IO: agent_costs[] (in/out/cache/reasoning tokens, elapsed_s, cost_usd)
  |- Reliab:  decision_attempts, parse_failures
  ~- Audit:   observation_hash, action_hash, prev_round_hash, tool_calls_log, signed
```

sentiment_trajectory, *candidate_signals*, the *mapc_delta_fields*, *agent_costs[]*, *parse_failures*, and *prev_round_hash** are the fields the **diagnostic scorecard** (§6.4) and the **hash chain** (§4.4) read. Crucially, every metric in this paper is recoverable from this record — no new instrumentation was added to compute the scorecard.

3.2 Six-Agent Architecture

CLQT employs a **six-agent architecture** where specialized agents receive **agent skills** — structured workflow protocols — that define the sequence of MCP tool calls and reasoning steps. Each agent (Research, Analyst, PM, Risk Officer, Execution, Reflection) maps onto a stage of the loop and is granted only the tool subset its role needs; this **stage-to-agent-to-tool mapping** is shown in Figure 2, and the exact skill files and tool assignments are available from the authors on request.

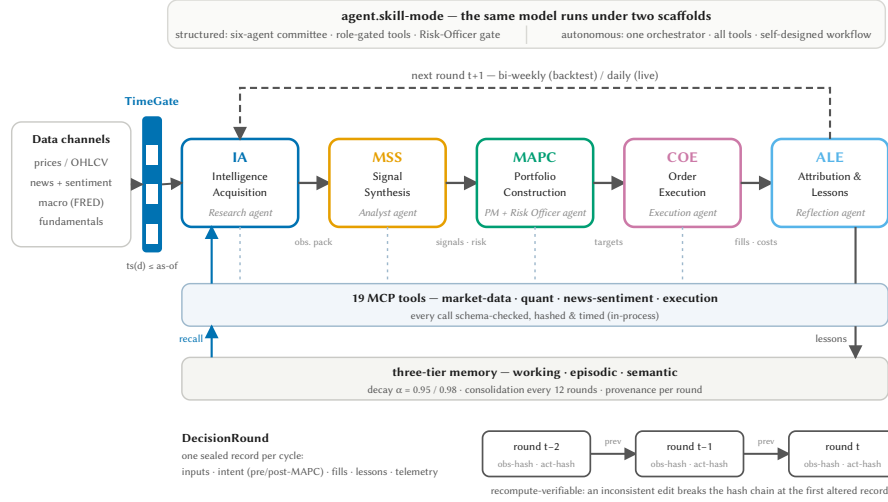


Figure 2. The CLQT closed decision loop. The five stages (IA $\rightarrow$ MSS $\rightarrow$ MAPC $\rightarrow$ COE $\rightarrow$ ALE) are staffed by six specialized agents in structured mode — a strict investment-committee process whose staged roles enforce the guardrails — or by a single full-access agent with end-to-end decision autonomy in autonomous mode. Each cycle reads from and writes to the three-tier memory and emits one DecisionRound: a sealed, hash-linked audit record.

Structured vs. Autonomous Skill Modes CLQT supports a **skill rigidity ablation** via *agent.skill-mode*:

- **structured:** Six specialized agents with restricted tool subsets matching their stage role and prescribed skill protocols. Agents reason *within* the workflow — formulating views, adjusting signal weights — but cannot change the workflow topology. A Risk Officer gate may reject and request a revision.
- **autonomous:** A single **AutonomousOrchestratorAgent** owns the entire pipeline from SCOUT through REFLECT, receiving all tools and only a mandate document, constraint set, and required output schema. It decides which tools to call, in what order, how often, and whether to skip stages.

Both modes use **real agentic tool calling** (function calling): LLM tool-use blocks are executed in-process against the actual quant and data libraries, with results fed back in a multi-turn loop. This is not simulated tool use or pre-assembled prompt data. (As §6.4/§7 detail, this mode switch is a confound for two axes — *Discipline* and *Acuity* — that we control by comparing within mode.)

3.3 Multi-Source Information Gathering and 3-Dimensional SCOUT Stage

Standard Data Channels The IA stage gathers data from complementary sources, each enforced through the **TimeGate** (§4.1) and each mapped to a backtest and a live provider so the two tracks see equivalent inputs (Table 2):

Data Channel	Fields	Backtest source	Live source
Price/Volume	OHLCV + adjusted close (252-day lookback)	Yahoo Finance	Yahoo Finance / broker snapshot
Fundamentals	P/E, EPS, revenue, earnings surprise, analyst targets	Finnhub	Finnhub / SEC EDGAR
News/Sentiment	Headlines, sentiment score	Alpha Vantage / Polygon / RSS / NewsAPI	same
Macro	12 series (GDP, CPI, unemployment, fed funds, VIX, ...)	FRED	FRED
Benchmark	SPY, IEF buy-and-hold NAV	Yahoo Finance	Yahoo Finance

Table 2. *Data sources and channel registry (backtest and live providers per channel).*

3-Dimensional SCOUT Stage Real universe candidates differ not only in momentum but in earnings quality and correlation structure with macroeconomic regimes. CLQT formalizes this through a **3-dimensional SCOUT stage** inserted between IA and MSS. For every candidate-pool symbol not currently held, the stage computes, using data available as of *as_of_ts*:

- **Dimension 1 — Technical Momentum** (m_i): 12-month minus 1-month price return.
- **Dimension 2 — Core Fundamentals** (f_i): most recent earnings-surprise percentage (Finnhub/EDGAR), with analyst-consensus deviation as fallback.
- **Dimension 3 — Macro Correlation** (ρ_i^{mkt}): 60-day rolling Pearson correlation of daily log returns with SPY; a companion ρ_i^{rate} (vs. TLT) captures rate sensitivity.

All three are z-scored cross-sectionally and combined:

$$z_i^{\text{composite}} = 0.50 \cdot z_i^{\text{mom}} + 0.30 \cdot z_i^{\text{eps}} + 0.20 \cdot z_i^{\text{mkt_corr}}$$

with a degenerate guard (correlation set to None and weight redistributed when the window has <10 observations or near-zero variance). The top- K candidates (default $K = 5$) are surfaced to MAPC and recorded in *candidate_signals*. The 50/30/20 split is an ablation default, not a derived optimum (*ablation_scout_weights* reserved). Figure 3 illustrates the ranking on a real campaign example (JNJ vs. FCX).

3.4 Cost-Aware Execution

CLQT tracks position-level PnL via **weighted-average cost basis**. On a buy of q_{fill} shares at p_{fill} into a position of q_{old} at $\bar{c}_{\text{old}}$:

$$\bar{c}_{\text{new}} = \frac{q_{\text{old}} \cdot \bar{c}_{\text{old}} + q_{\text{fill}} \cdot p_{\text{fill}}}{q_{\text{old}} + q_{\text{fill}}}$$

On a sell, $\text{PnL}_{\text{realized}} = (p_{\text{fill}} - \bar{c}) \cdot q_{\text{fill}} - c_{\text{tx}}$. **Daily financing costs** accrue for leveraged/short positions; the margin-interest leg prices off the round’s gathered FRED Fed-Funds series (5% fallback when the macro module is ablated) plus a 150 bps broker spread — under the default long-only, leverage- ≤ 1 mandate both financing legs are

3-Dimensional SCOUT: composite z-score ranking of universe candidates

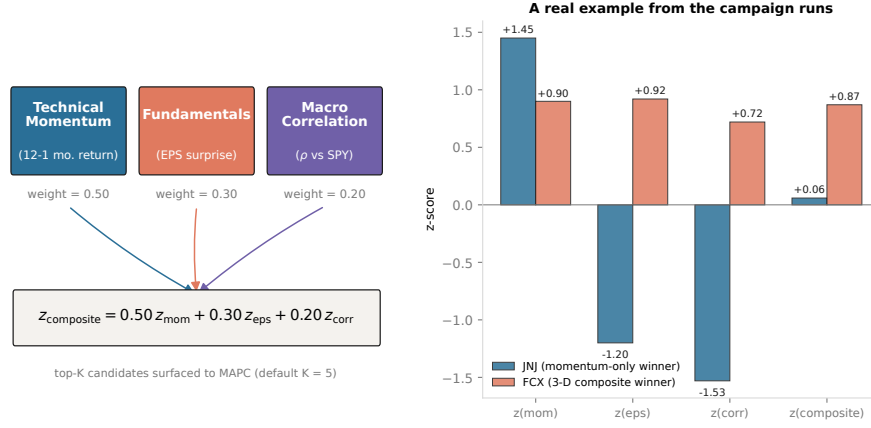


Figure 3. The 3-D SCOUT stage ranks universe candidates on a composite z-score of momentum, earnings surprise, and macro correlation. The right panel is a real example from the campaign runs: JNJ leads on momentum alone, but a stronger earnings surprise and higher market participation make FCX the 3-D composite winner.

structurally zero. **Four market-impact models** (fixed BPS, square-root, Almgren–Chriss, structural) are selectable per **cost tier** (Appendix A, Table A1). The cost tier is experiment-configurable (not hard-coded), enabling the **cost-tier sensitivity ablation** (§7.7); the default is LOW, whose commission and spread legs are validated against live broker paper fills (§5.2); its impact leg is parameterized (square-root), not fitted to fills, since paper-account fills embed no market impact.

3.5 Strategy Consistency Measurement

CLQT computes a **ConsistencyScore** $\mathcal{C}_t$ after each round:

$$\mathcal{C}_t = 0.25 \cdot (1 - d_{\text{style}}) + 0.25 \cdot \tau_{\text{consist}} + 0.25 \cdot s_{\text{adhere}} + 0.25 \cdot a_{\text{target}}$$

capturing **style drift**, **turnover consistency**, **signal adherence**, and **mandate alignment**. When $\mathcal{C}_t < 0.7$ a **drift warning** is injected with a per-asset delta cap $|\delta_i| \leq \bar{\delta}/2$. The components feed the Composure and Discipline axes (§6.4); Figure 4 shows the four-component decomposition.

3.6 Structured Multi-Modal Memory System

A **three-tier memory architecture** (Figure 5) accumulates and generalizes behavioral patterns across sessions:

- **Working memory:** current observation pack plus the most recent $N = 3$ rounds.

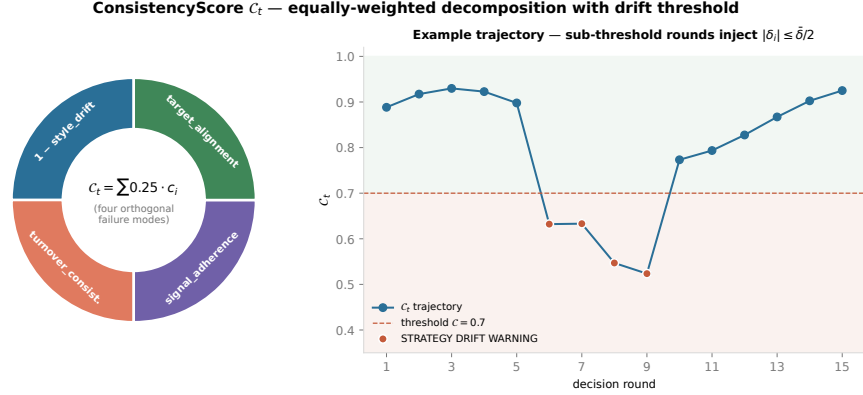


Figure 4. *ConsistencyScore decomposition into its four orthogonal components and the drift-warning threshold.*

- **Episodic memory:** structured event-action-outcome records, decaying at $\alpha_e = 0.95/\text{round}$.
- **Semantic memory:** generalized cross-asset patterns consolidated from episodic records, $\alpha_s = 0.98$.

Every $k = 12$ rounds a consolidation pass merges episodic records sharing asset/event/outcome patterns. *memory_provenance* records which memories influenced each round, enabling the memory ablation (§7.7). Decay is applied at retrieval time as $\alpha^{\Delta_{\text{round}}}$ and contributes the recency term of the relevance score (alongside asset/event/regime overlap). The two rates are **fixed defaults, not tuned against returns** — they encode only the prior that specific episodic records should fade faster than generalized semantic patterns; $\alpha_e = 0.95$ and $\alpha_s = 0.98$ per round correspond to half-lives of ≈ 13.5 and ≈ 34 rounds, and the episodic rate is exposed as a config parameter for tuning or ablation.

3.7 Knowledge Synthesis and Decision Pipeline

The **MAPC decision** reconciles a **quantitative optimizer proposal** with the **PM agent’s tactical overlay** through four layers: **quantitative proposal** (w_{opt}), **budget-neutral agent overlay** (δ with conviction scores), **constrained projection** onto the feasible set $\mathcal{F}$, and **Risk Officer validation** with an execution gate. The agent’s proposed overlay δ is recorded *before* projection as *mapc_delta_proposed*, and the projected result as *mapc_delta_projected*, with *mapc_adjustments_clipped* counting constraint clips. This separation is what lets the scorecard measure an agent’s *intent* (Acuity, Coherence on proposed deltas) distinctly from the optimizer/MAPC machinery that may overwrite it.

Dynamic Universe Management The top- K **SCOUT candidates** are surfaced to the PM alongside current holdings, enabling **agent-driven rotation**, constrained by a

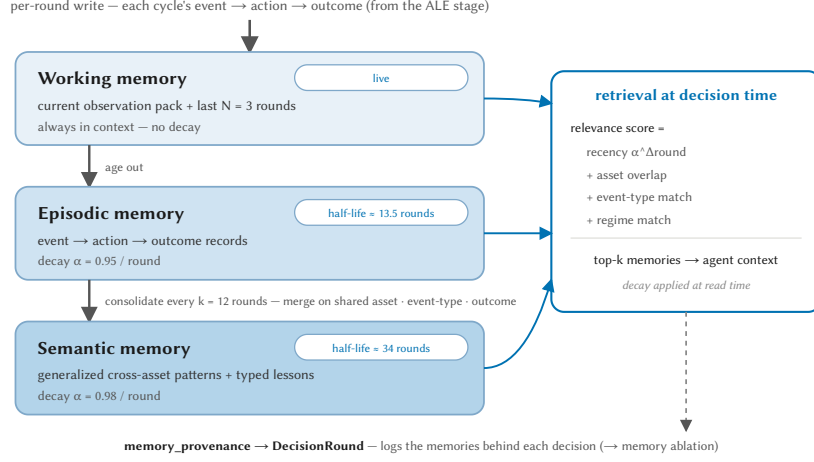


Figure 5. Working / episodic / semantic memory tiers with decay rates and the periodic consolidation pass.

per-round addition cap, a minimum holding-period guard, and the turnover budget. A **post-parse cross-validation pass** resolves logical contradictions.

3.8 Intertemporal Sentiment Trajectory and News Persistence

Prior work treats news as a point-in-time context. CLQT instead surfaces *how* sentiment is evolving. `gather_news()` uses a **dynamic lookback** $L_t = \max(7, t_{\text{as_of}} - t_{\text{last_rebalance}} + 1 \text{ day})$ and a persistent **NewsStore** that deduplicates and supplements articles across runs, so weekend or prior-round articles remain available within free-tier API limits. Articles are grouped into per-(symbol, day) **SentimentDay** objects (mean sentiment, count, most-extreme headline) forming a **sentiment trajectory**, classified IMPROVING/DETERIORATING/STABLE by comparing the last-3-day to first-3-day mean against $\delta_{\text{thresh}} = 0.1$. The PM receives a structured trajectory table (Figure 6); the trajectory is stored in `DecisionRound.sentiment_trajectory` for replay. The *news-sentiment* module ablation (§7.7) toggles this channel.

4. Data Integrity

A closed-loop trading number is only as trustworthy as the data discipline behind it — apparent agent alpha can largely dissolve once look-ahead bias is controlled [8, 9]. CLQT therefore treats two guarantees as preconditions rather than features: no decision may see the future (§4.1–§4.3), and the record of what each decision saw and did must be independently checkable (§4.4).

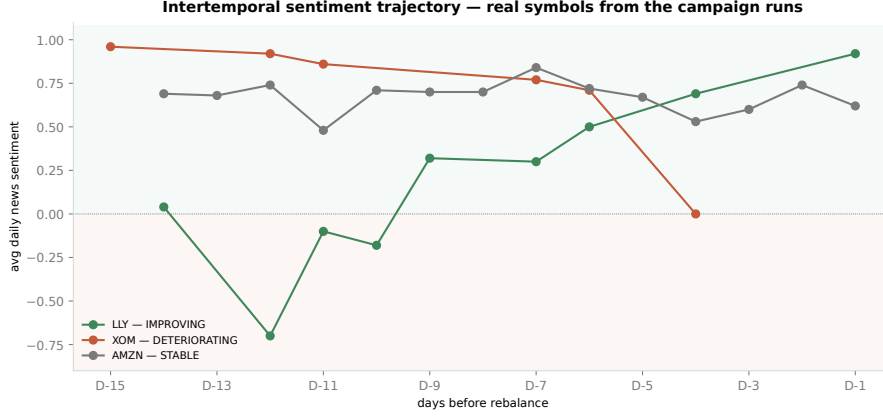


Figure 6. Per-symbol intertemporal sentiment trajectory across the inter-rebalance window, with the improving / deteriorating / stable trend classification surfaced to the PM agent. Three real symbols from the campaign runs: LLY (improving), XOM (deteriorating), and AMZN (stable).

4.1 TimeGate: Point-in-Time Enforcement

All data access passes through **TimeGate**, enforcing $\forall d \in \mathcal{D}_t : \text{timestamp}(d) \leq t_{\text{as_of}}$. For live mode an *intraday_cutoff* freezes access N minutes before the decision timestamp. The TimeGate applies uniformly to all modalities, including memory retrieval and the **NewsStore** supplement pass.

4.2 Contamination Prevention

Post-cutoff windows. Per-model eligibility filtering ensures backtest data postdates each model’s training cutoff. Universe integrity. Point-in-time universe membership avoids survivorship bias.

4.3 Contamination Detection

Two complementary probes are implemented: canary documents (synthetic disclosures with unique identifiers detected via substring + fuzzy match) and future-shift leakage probes (paired baseline/probe decisions with normalized L1 weight divergence). These detection probes are part of the substrate but were not exercised in this single, benign campaign; contamination resistance here rests on the point-in-time gate and per-model cutoff filtering (§4.2), the live post-cutoff track (§7.10), and the verified hash chain (§4.4). Active adversarial/canary probing to stress the capability axes directly is a hardening direction (§8.8).

4.4 Recompute-Verifiable Audit Trail

Each **DecisionRound** is sealed into a **recompute-verifiable hash chain**: a content hash over the inputs the agent saw and a content hash over the decision it took, with each

round’s hash folded into the next. Verification re-derives every hash from the persisted record and checks both that it matches the stored value and that the chain linkage holds. An inconsistent edit — a changed weight, a deleted order, or a reordered round without recomputing its successors — breaks the chain at the first altered record. This claim is deliberately scoped: the chain checks the integrity of a provided store; it is not an externally anchored cryptographic signature and does not provide non-repudiation against wholesale replacement and recomputation of the store. Every campaign run passes this end-to-end integrity verification.

5. Backtest and Live Evaluation Protocols

With the substrate (§3) and its integrity guarantees (§4) in place, this section defines the two tracks that generate the paper’s evidence — a contamination-aware backtest (§5.1) and a broker paper-trading live track (§5.2) — together with the rebalancing semantics (§5.3) and the protocol for comparing across tracks (§5.4).

5.1 Backtest Track: Contamination-Aware Period Design

Rather than a single fixed period, CLQT assigns models to evaluation groups by training cutoff: Group B (cutoff > 2 years ago) → 2-year backtest from the cutoff; Group A (cutoff > 1 year ago) → 1-year backtest. Backtest parameters: 252-day warmup; configurable initial capital (default \$1,000,000); bi-weekly rebalance (the 26-round campaign cadence, §7.2); LOW cost tier by default; deterministic given the configuration. In this paper’s campaign every evaluated model’s cutoff precedes the single 1-year window (§7.2), so the longer 2-year Group B path, though supported, is not exercised by this cohort.

Universe and starting book. The investable universe is derived from the S&P 100 — large, liquid, survivorship-controlled names with point-in-time membership (a symbol enters only once it was actually in the index as of the decision date). Each run opens from a core book of roughly 14 equal-weighted holdings carried as the starting portfolio, against a candidate pool of ~58 additional names (≈ 72 symbols in total) that the SCOUT stage (§3.3) ranks for rotation. The universe is deliberately broad enough to permit meaningful selection and turnover, yet bounded so the cohort stays comparable across models and the news/fundamentals channels remain tractable within free-tier API limits.

5.2 Live Track: Broker Paper-Trading Architecture

A live track serves two purposes: (1) genuine contamination resistance — decisions execute on data that postdates every model’s training cutoff — and (2) a check that the benchmark’s execution abstractions map onto a real broker’s order lifecycle. CLQT integrates a commercial broker’s paper-trading API. Each session gathers and decides with the decision timestamp fixed before that day’s market open — a Monday session’s gather window covers the weekend, and the same pre-open as-of convention as the backtest keeps the two tracks comparable; it then previews and submits orders at the open, polls fills, reconciles the resulting book from the broker’s positions and cash ledger, and locks the portfolio between rebalances. A dedicated execution agent optimizes only cost and fill quality — the target weights are fixed by the upstream decision. The live

track thus executes real fills on genuinely unseen, post-cutoff data; the broker integration is documented and available from the authors on request.

5.3 Rebalancing vs. Re-Optimization

CLQT distinguishes rebalancing (mechanical weight restoration) from re-optimization (full pipeline producing a new target) via three rebalance modes — calendar, threshold, and agent-discretion — enabling evaluation of *when-to-trade* judgment.

5.4 Cross-Track Comparison Protocol

When the two tracks share a window and configuration, they are reported on the same metric suite so discrepancies are attributable: a backtest Sharpe exceeding live by more than ~ 0.2 flags possible contamination, while fill-price divergence between simulated and live execution gauges cost-model calibration (the fill-price leg is how the LOW cost tier was calibrated, §3.4). In this paper’s campaign the two tracks deliberately differ — a year of bi-weekly backtest rounds versus four weeks of daily live rounds on a newer model generation — so the cross-track comparison runs at the *decision* level on track-invariant axes rather than on returns. The live campaign itself — four trading weeks, eighteen valid NYSE days, now complete — is reported in §7.10, after the evaluation framework (§6) has defined the axes it is read with.

6. Evaluation Framework

The preceding sections built the system that emits the audit trail; this section defines what is measured from it. We describe the ablation dimensions the experiments will toggle (§6.1), the passive suite every run is scored against (§6.2), the recorded metric dictionary (§6.3), the five-axis capability scorecard that is the benchmark’s central read-out (§6.4) with its held-out judge protocol (§6.4.1), and the behavioral-interpretability method that complements it (§6.5).

6.1 Ablation Dimensions

CLQT supports 12 ablation dimensions in three categories (Table 3). Module flags: *news-sentiment*, *world-macro*, *cost-modeling*, *advanced-quant*, *systematic-execution*, *reflection*, *memory*, *investment-target*, *asset-class-data*. Config overrides: *cost-tier* $\in$ {zero, low, medium, high}, *rebalance-mode* $\in$ {calendar, threshold, agent-discretion}. Process autonomy: *skill-mode* $\in$ {structured, autonomous}.

Group	Dimensions	What it isolates
Information	<i>news-sentiment</i> , <i>world-macro</i> , <i>asset-class-data</i>	what data the agent can access
Decision quality	<i>advanced-quant</i> , <i>investment-target</i>	optimization and mandate enforcement
Execution	<i>cost-modeling</i> , <i>systematic-execution</i>	trade costing and execution strategy
Learning	<i>reflection</i> , <i>memory</i>	whether the agent learns across rounds
Regime	<i>cost-tier</i> , <i>rebalance-mode</i>	environmental conditions
Autonomy	<i>skill-mode</i>	process prescription vs. agent-designed workflow

Table 3. *Ablation dimension taxonomy.*

6.2 Baselines and Benchmark Comparison

The instrument scores agents against a full passive suite that every run computes automatically: the index ETFs SPY (S&P 500 proxy) and IEF (7–10y Treasuries); a synthetic 60/40 ($0.6 \cdot \text{SPY} + 0.4 \cdot \text{IEF}$); an equal-weight portfolio of the agent’s own investable universe (EW-Universe); and four algorithmic baselines — buy-and-hold, equal-weight (rebalanced), risk-parity (inverse-vol), and 12-1 momentum. For each, the run reports cumulative/excess return, information ratio, and tracking error. The headline question — *does any agent beat the cap-weighted index net of costs?* — is answered against this suite, not a single benchmark.

6.3 Metric Dictionary

The instrument records a broad metric suite — performance, benchmark-relative, cost, governance, consistency, learning, efficiency, reliability, and integrity — all recoverable from the audit trail (Table 4).

Category	Metrics
Performance	CumRet, AnnRet, Vol, Sharpe, Sortino, Calmar, MaxDD
Benchmark-relative	excess, IR, TE vs SPY/IEF/60-40/EW-Universe + 4 baselines; “beat /8”
Cost-awareness	turnover, bps decomposition (spread/commission/slippage/impact), net-vs-gross, financing drag
Governance	RO-reject %, no-trade %, MAPC clips/round, constraint violations
Consistency	composite C, style drift, turnover consistency, signal adherence
Closed-loop learning	lessons/round, intra-run learning curve
Efficiency	\$/round, in/out/reasoning tokens, latency, \$ per 1% excess
Reliability	completion, parse-fail, timeout, output-cap-hit
Integrity	chain-verify pass, leakage divergence, canary recall, cutoff compliance

Table 4. *Metric dictionary (all recoverable from the audit trail; cf. Appendix D). Sharpe and Sortino are computed against a zero risk-free rate (ratios of annualized return to annualized total/downside volatility), the convention in the agent-trading benchmarks we compare against [1, 2, 6]; with 2025–26 cash yields near 4–5% and a mandated $\geq 2\%$ cash buffer, subtracting a risk-free leg would shift all agents’ ratios down by a similar amount without changing rankings.*

6.4 The Five-Axis Diagnostic Capability Scorecard (APM-CS)

The scorecard is the instrument’s read-out. It scores five capability axes from the audit trail, ordered high→low by how fundamental they are to competent autonomous portfolio management. Each axis is a finance instantiation of a general agent-evaluation capability, so a finding about a trading agent generalizes to a statement about the model’s reasoning. Outcome (returns) is *not* an axis — it is a result explained by the axes.

D1 • Coherence — *general capability*: global, systems-level reasoning vs. local greedy pattern-matching; an analogue of systematic alignment vs. turn-by-turn instruction-following at the level of reasoning (does the conclusion follow from the agent’s own analysis?). *Finance*: the chosen allocation follows from the agent’s stated research and

the signals it cited. *Synthesis*:

$$D_1 = 0.5 A_{\text{sig}} + 0.5 J_{\text{judge}}$$

where A_{sig} is signal-action agreement: the fraction of the agent’s proposed trades (*mapc_delta_proposed*, pre-MAPC) whose direction matches the sign of that name’s cross-sectional informative composite (below). J_{judge} is the held-out LLM judge’s score (§6.4.1), normalized to $[0,1]$. Reveals shallow or self-contradictory reasoning that returns hide.

D2 • Acuity (*information attention / acuteness*) — *general*: salience attention vs. distraction, i.e. endurance under polluted / misleading context. *Finance*: attend to informative signals amid a large noisy universe and a flood of news. *Synthesis*, on proposed adjustments (not final weights, which are confounded by the optimizer/MAPC):

$$D_2 = \rho_s(\Delta_{\text{prop}}, I) - \rho_s(\Delta_{\text{prop}}, N)$$

where ρ_s is Spearman correlation, $I = z(m_{12-1}) + z(e_{\text{surp}}) + z(n_{\text{sent}})$ is the informative composite, and $N = \text{mean}(z(\sigma_{\text{realized}}), z(\text{RSI}_{14}))$ is the two-leg noise composite (v1.1): volatility level, which carries no directional alpha at this horizon, and the 14-day RSI oscillator, which is salient in the data feed but horizon-mismatched to a bi-weekly rebalance. Short-term *reversal* is deliberately excluded — it is a documented (if weak) alpha factor, and penalizing attention to it would mislabel signal as noise. The single-leg (realized-vol-only) variant is retained as a robustness column (Appendix F.1b). Each term is z-scored cross-sectionally per round and averaged over rounds with ≥ 3 proposed trades. Positive $\Rightarrow$ the agent’s intent tracks alpha-bearing signals more than the distractors; near-zero or negative $\Rightarrow$ chasing noise.

D3 • Composure — *general*: robustness / consistency under input perturbation vs. reactive overcorrection. *Finance*: respond to volatility and misleading data with discipline, not churn. *Synthesis* (the mean of its legs, not a product):

$$D_3 = \text{mean}(1 - s_{\text{react}}, c_{\text{cons}})$$

where s_{react} is the (squashed, non-negative) regression coefficient of per-round one-way turnover on a market-volatility proxy (cross-sectional mean *realized_vol*), and c_{cons} is the strategy-consistency leg when available; a high slope means turnover rises sharply with volatility. We score Composure as the average of its legs rather than their product for two reasons made concrete by the campaign: a product lets one weak leg collapse the whole axis (it sent a single model’s Composure to ~ 0), and — because the *strategy-consistency* leg is produced by the memory module — a product made Composure uncomputable under the memory ablation (§7.7). The mean is robust to a single low leg and falls back to the memory-free *calm* leg when *consistency* is absent, so Composure is defined for *every* configuration including memory-off ablations. (Style-drift is ≈ 0 cohort-wide, so it is reported separately rather than carried as a non-discriminating leg.) Reveals models that overreact to volatile or misleading inputs.

D4 • Discipline — *general*: systematic alignment to a standing mandate vs. turn-by-turn instruction-following — does the agent self-regulate to a policy even absent a hard

guardrail? *Finance*: respect caps/turnover/risk budgets *without* relying on the MAPC clip. *Synthesis* (bounded to $[0, 1]$):

$$D_4 = \frac{1}{2} r_{\text{self}} + \frac{1}{2} (1 - d_{\text{clip}})$$

where r_{self} is the self-constrained rate: the fraction of rounds whose target satisfies constraints pre-MAPC. $d_{\text{clip}} = \min(1, \bar{a}_{\text{clip}}/\kappa)$ is clip dependence, measuring reliance on the reconciler to *force* the book inside the mandate ($\kappa = 10$, a fully-saturated structured round; $\bar{a}_{\text{clip}}$ is the mean number of clipped MAPC adjustments). Both legs lie in $[0, 1]$, so Discipline does too. (A turnover-cap leg used in earlier drafts was dropped: campaign turnover ~ 0.15 sits far below any real cap, so the leg was a non-discriminating constant.) *Mode confound*: RO-approval is meaningful only in *structured*; *autonomous* sets *risk_approved*=*True* by construction — so Discipline is compared within mode.

D5 • Reliability — *general*: protocol/format adherence + operational robustness + efficiency — the analogue of an agent completing a tool-use task in valid form, every time, at reasonable cost. *Finance*: close the loop in valid schema at acceptable compute. *Synthesis*:

$$D_5 = r_{\text{complete}} (1 - r_{\text{parse}}) (1 - r_{\text{trunc}})$$

where r_{complete} is completion rate, r_{parse} is parse-fail rate, and r_{trunc} is truncation rate. Efficiency is reported beside it (\$/round, reasoning-tokens/round, latency). As of v1.1, truncation is detected against the *role-level* output cap recorded on each call (e.g. the PM’s 2,048-token budget) rather than the global 8,192 budget alone — a PM decision truncated at its role cap is exactly the failure class this axis exists to count, and was previously invisible to it; the completion leg is clamped at 1 so an under-specified round plan can never reward a run with Reliability > 1 .

Scoring & aggregation. Each axis is mapped to a cohort percentile and the five form a radar fingerprint (Figure 7); Table 5 tabulates the percentile profile and a composite APM-CS (mean of the five percentiles). Pairwise dominance and win-rate across (axis $\times$ sub-period $\times$ run), with rank-stability (Kendall’s τ), are planned broader-run outputs (§8.8). The thesis bar — *superior* = *consistent dominance across axes and sub-periods* — requires a configuration to win the large majority of axes against all rivals and sustain that pattern across regime sub-periods and runs. Because D2 and D4 are mode-confounded, their raw values are compared only within mode; cross-mode APM-CS contrasts are descriptive profile summaries, not clean estimates of a scaffold effect. APM-CS is also cohort-relative: adding or removing configurations rescales it. The raw D1–D5 values are therefore the portable read-out.

6.4.1 Held-out LLM-as-Judge for Coherence The judge component of Coherence is scored by a held-out, non-cohort model — i.e. a model *not* in the evaluated set — to avoid the self-preference bias documented for LLM-as-a-judge. For each round the judge receives a compact, audit-derived prompt: the agent’s stated analysis (research brief and/or optimizer rationale), the top names by informative-composite strength with their underlying signals, the chosen target weights, and the proposed weight changes. It returns a 1–5 score for the single question “*does the chosen allocation follow coherently from the stated analysis and signals?*” — explicitly told to ignore profitability and to

penalize self-contradiction and untethered allocations. Scores are cached by *round_id* (so re-runs never re-bill) and reasoning is enabled on the judge model deliberately: coherence adjudication is nuanced and the held-out judge is a reasoning model, so we trade a few cents per round for adjudication quality. Bumping the prompt version invalidates the cache so all rounds are re-judged consistently when the rubric changes.

Two-judge protocol (per-track primaries + cross-corroboration). *Held-out* means held out of the cohort actually being scored, and as of v1.1 this is enforced in tooling: the scorecard builder refuses to run when the judge model matches any scored run (an explicit debug override exists but taints the affected rows by construction). Since minimax-m3 sits in the *live* panel, the two tracks use different primaries — **minimax-m3 judges the backtest cohort** (it is not in it) and **GLM-5.2, a different-family reasoning model outside both cohorts, judges the live track**. The two judges are calibrated against each other on the backtest rounds they share (Spearman $\rho = 0.65$, 97 % within one point on the 1–5 rubric), and the agreement $\leftrightarrow$ judge coherence gap (§7.10) — the instrument’s strongest claim — reproduces under a full judge swap on the backtest (§8.7), so the headline finding is a property of the allocations, not of one judge. Neither judge’s scores are blended with the other’s; each track reports its own primary.

6.5 Behavioral Interpretability: Method

The same audit trail that powers the quantitative scorecard (§6.4) also supports a qualitative, behavioral read — extracting each model’s decision behaviors, stated reasoning, and recurring patterns into a human-legible picture of its “decision personality.” This is the tradition of work that probes LLM black boxes through their *observable actions and self-reports* rather than their weights, and CLQT is unusually well-suited to it because every round is a complete, hash-chained, verifiable record (§4.4). Captured per round, per model $\times$ mode, with no new instrumentation (the exact audit fields behind each are mapped in Appendix D, Table D1):

- **Stated reasoning** — the agent’s own narrative account of *why* an allocation was chosen: research and analyst briefs, the allocation rationale, risk-officer notes, and the post-round reflection.
- **Metacognition and learning** — how a model abstracts experience into typed, reusable “lessons”, and whether those lessons are later retrieved back into context.
- **Intent vs. guardrails** — the gap between what the model *proposed* and what the reconciler *allowed* (in a sampled structured run the agent proposed 12–14 weight moves per round and 7–12 were clipped — a quantifiable “intent meets institutional guardrails” signal).
- **Inter-agent dynamics** — how the PM and Risk Officer negotiate, and when the committee overrides itself.
- **Information foraging** — each model’s characteristic data-gathering strategy: which tools it calls, in what order, and how widely it explores the candidate universe.
- **Cognitive effort** — where a model “thinks hardest” (reasoning-token and latency budget per role) and whether that effort tracks decision quality.
- **Struggle signatures** — where and why a model fails to produce a valid decision:

retry loops, parse failures, and constraint violations.

Method. Thematic coding of the reasoning corpus (optionally LLM-assisted via a *held-out* summarizer mirroring the judge protocol in §6.4.1), frequency analysis of intent-vs-clip and committee overrides, per-model tool-use fingerprints, lesson formation and reuse, and illustrative anecdotes — each tied to a hash-verified round so it is reproducible.

Scope (honest caveat). We analyze stated and observable reasoning artifacts and behavioral metadata, not raw hidden chain-of-thought: providers generally do not return the thinking trace, and we persist reasoning-token *counts*, not the trace text. The interpretability is therefore of *behavior + self-report* — auditable and reproducible (every claim pinned to a hash-chained round), but situated one layer above the model’s hidden cognition. We treat this scoping as a feature, not a gap: it keeps the “human reading of the AI” grounded in verifiable evidence rather than unfalsifiable introspection.

7. Experiments

The experiments here are the full 1-year campaign — still not a model ranking, but a *map of limitations* across a 5-model cohort. The purpose is to show that the five-axis scorecard discriminates decision *quality* where outcome metrics conflate it, surfaces model-specific failures invisible to returns, and — by the instrument’s own standard (*consistent dominance across axes and sub-periods*) — resists premature ranking even with repeated-run evidence. Rows report means over repeated runs; bootstrapped CIs and Kendall- τ rank-stability are a planned broader-run pass (§8.8). The method, the axes, and the audit machinery are the contribution.

7.1 Research Questions, Answers, and Where the Evidence Lives

So the reader can navigate the campaign with its conclusions in hand, each research question is stated here with its short answer and the section that carries the evidence.

- **RQ1 (instrument validity):** Does the scorecard discriminate decision *quality* where outcome metrics conflate it? — **Yes.** The scorecard separates configurations that returns conflate: the nominal Sharpe “winner” carries the cohort’s lowest composite and a Reliability of 0.19 (§7.3-F1), and module removals that leave Sharpe flat register plainly on the capability axes (§7.7).
- **RQ2 (capability $\neq$ outcome):** Is the Sharpe leader the capability leader, and does any configuration dominate all five axes? — **No and no.** The capability leader (deepseek · structured, APM-CS 72.9) is mid-pack on Sharpe, and no configuration encloses all five axes (§7.3-F1, §7.4) — which is why, by the instrument’s own standard, the cohort is not cleanly rankable.
- **RQ3 (model-specific limitations):** Does the audit trail surface model-specific failures invisible to returns? — **Yes.** Three distinct backtest reliability deficits with three different mechanisms (§7.5), and two distinct live hold mechanisms that a targeted scaffold fix then eliminated (§7.11).
- **RQ4 (scaffold effect):** How does the structured $\leftrightarrow$ autonomous mode shape the capability profile? — **Model- and axis-dependently.** The backtest composite descriptively favors structure for the reliable configurations while the live cohort

uniformly favors autonomy; the mode-comparable sensitivity reconciles the two reads (§7.6, §7.10, Appendix F.4).

- **RQ5 (module importance):** Can module value be ranked from returns? — **No, not even with repeated runs.** Every single-module and cluster knockout lands inside the ± 0.42 repeated-run noise band; what a module contributes registers on the capability and behavioral axes instead (§7.7).
- **RQ6 (cost realism):** Are the cost tiers calibrated? — **LOW yes, HIGH no.** LOW’s commission and spread legs validate against live broker fills, while the HIGH tier’s implausible per-round drag exposes its own mis-calibration — the instrument auditing itself (§7.7).
- **RQ7 (passive comparison):** Net of costs, does any agent beat the passive suite? — **The defensive baselines yes, the cap-weighted index not cleanly.** Reliable structured configurations clear a median ~ 6 of 8 baselines; the index shortfall reads as a deliberate de-risking trade, not absent skill (§7.9).

7.2 Setup

Campaign grid. Five models spanning families, sizes, and reasoning styles — qwen3-235b (large MoE, non-thinking), gemini-2.5-flash-lite (small), deepseek-chat-v3.1 (mid, hybrid→non-thinking), gpt-5-mini (reasoning, effort low), claude-haiku-4.5 (small, thinking off) — each in structured and autonomous skill modes, five repeated seeds per structurally-reliable configuration (gemini, deepseek, gpt-5-mini, haiku in both modes, and qwen3 autonomous; means over seeds; qwen3 · structured is a single showcase seed — seven seed attempts all failed to complete, §7.5). Period: 26 bi-weekly rounds, 2025-06-16 → 2026-06-12; S&P-100-derived universe (14 core + 58 candidate = 72), all modules enabled, LOW cost tier, initial NAV \$1M, context capped 32k for cohort comparability, uniform output budget 8192. Coherence judge: held-out, non-cohort minimax-m3, reasoning enabled, cached.

Cutoff safety. Every model’s training cutoff precedes the 2025-06-16 start (gemini Jan’25, gpt-5-mini May’24, claude-haiku Feb’25, deepseek Mar’25); qwen3 (Jun’25) sits within a documented ≤ 1 -month grace. The *IntegrityChecker* records cutoff compliance per run.

Ablation grid. Single anchor = gemini-2.5-flash-lite, structured, LOW (cheap; its structured runs complete): six module-off variants (*cost-modeling*, *memory*, *reflection*, *news-sentiment*, *world-macro*, *advanced-quant*), each as repeated runs, + two cost-tier variants (*medium*, *high*, single-run) — every run complete via *run_ablation.py* (§7.7, Figure 9).

Parsing hardening. Before the campaign, the PM/Risk schema parsers were hardened to drop null list items (and lift bare-string items) in *adjustments/checks* — small models (notably gemini in structured mode) emit `{“adjustments”: [null, ...]}`, which previously failed validation and degraded the round. This recovers the valid content and materially improves structured Reliability across the cohort (§7.5).

7.3 Campaign Results (RQ1, RQ2)

Table 5. *Per (model × mode), 26 bi-weekly rounds, hash-chains verified, sorted by APM-CS. Means over **five seeds** per configuration (qwen3 · structured: single showcase seed — see note). Beat /8 vs {SPY, IEF, 60/40, EW-Universe, buy-and-hold, equal-weight, risk-parity, momentum}. D1 Coherence = 0.5 · agreement + 0.5 · held-out minimax-m3 judge (the backtest primary, §6.4.1); D2 Acuity uses the v1.1 two-leg noise composite (§6.4); D5 Reliability = completion · (1−parse-fail) · (1−truncation), role-cap aware. APM-CS = mean cohort percentile across D1–D5. Axes D2/D4 are mode-confounded — compare within mode. Bold marks each column’s most diagnostically notable value(s) — the cohort best (Acuity, APM-CS) or the notable floor/artifact the analysis turns on (the two Sharpe outliers; haiku’s Coherence floor; gemini-autonomous’s Composure floor; the gemini/deepseek-autonomous/qwen3 Reliability deficits; qwen3’s negative Acuity) — rather than uniformly the per-axis leader.*

Model · mode	Sharpe	CumRet	Beat/8	D1	D2	D3	D4	D5	APM-CS
deepseek-chat-v3.1 · structured	2.99	+28.6%	6.8	0.45	0.07	0.93	0.90	0.99	72.9
claude-haiku-4.5 · structured	3.21	+23.4%	6.0	0.23	0.44	0.95	1.00	1.00	71.0
gpt-5-mini · structured	2.62	+22.5%	4.6	0.43	0.08	0.92	0.79	1.00	67.4
gpt-5-mini · autonomous	1.76	+22.2%	4.6	0.47	0.03	0.92	0.42	0.99	50.3
gemini-2.5-flash-lite · structured	2.31	+24.8%	6.0	0.38	0.07	0.93	0.84	0.61	47.4
claude-haiku-4.5 · autonomous	1.92	+22.5%	4.8	0.44	0.01	0.81	0.61	0.98	40.4
qwen3-235b · autonomous	1.20	+12.6%	2.2	0.42	−0.06	0.91	0.56	0.99	39.9
deepseek-chat-v3.1 · autonomous	1.67	+21.5%	4.4	0.42	0.00	0.90	0.64	0.62	35.1
gemini-2.5-flash-lite · autonomous	1.75	+15.5%	3.0	0.47	−0.07	0.77	0.47	0.95	32.3
qwen3-235b · structured ★	9.82	+13.1%	8.0	0.33	−0.79	0.95	0.80	0.19	30.7

★ **qwen3-235b structured** is reported on its completed rounds only — seven seed attempts never finish more than 10 of 26 rounds (see §7.5). The five-seed extension root-caused this as tool-loop non-termination inside the structured committee (an agentic orchestration gap; earlier truncation and rate-limit hypotheses were symptoms), while the *same* model completes **autonomous** runs cleanly — five complete seeds — at the same context cap and toolset. Its structured Sharpe (9.82) / beat (8/8) are therefore on a *5-round self-selected window* and are meaningless except as a foil — read them only against its Reliability (~0.2). The autonomous rows (five seeds) are the trustworthy qwen3 rows.

Findings.

- **F1** — Capability leader ≠ outcome leader (the thesis, in one table). deepseek · structured tops APM-CS (72.9; the most balanced profile: D1 0.45, Discipline 0.90, Reliability 0.99), with haiku · structured close behind (71.0) on a very different shape — yet deepseek is *mid-pack* on Sharpe (2.99). The nominal Sharpe “winner” is qwen3 · structured at an implausible 9.82 / beat 8-of-8 — but on 5/26 rounds with Reliability 0.19, the lowest APM-CS in the cohort (30.7): the scorecard flags it as a self-selected-window artifact, not a strategy. Among *reliable* configs the Sharpe leader is haiku · structured (3.21), whose profile is the cohort’s most instructive split: the best Acuity point estimate (0.44 — its proposed trades track the informative signals over both distractors, though on a

thin sample; see F4) and perfect Discipline/Reliability, yet the *lowest* Coherence (0.23) — it points at the right signals but its allocations do not follow its own stated analysis. A leaderboard would crown qwen3 (a spurious 5-round Sharpe); the scorecard instead returns a *profile* per config and explains why the headline number misleads (Figure 7, Figure 14).

- **F2** — Scaffold effects are model- and axis-dependent. Under the full cohort-relative composite, structured APM-CS is higher for the four reliable multi-mode backtest models: deepseek 72.9 vs 35.1, haiku 71.0 vs 40.4, gpt-5-mini 67.4 vs 50.3, and gemini 47.4 vs 32.3. This is a descriptive profile contrast, not a clean mode effect, because D2 and D4 are mode-confounded; qwen3’s structured row also contains only five reliable rounds. The axis-level result is more informative: structure broadly raises Composure and contains particular reliability failures, but its value depends on the model and capability being examined. The live cohort favors autonomy more consistently (§7.10, Table 10), so committee scaffolding is not universally beneficial. A sensitivity restricted to the mode-comparable axes (the D1/D3/D5 mean, Appendix F.4) sharpens the point: on it, structured is favored only for deepseek in the backtest cohort, and all six live pairs favor autonomy.
- **F3** — Reliability deficits hidden by returns. *gemini · structured* posts a healthy +24.8% / Sharpe 2.31 yet Reliability 0.61 — residual schema parse-fails persist *even with retries=3*. *qwen3 · structured* completes only 5/26 rounds on its showcase seed (Reliability 0.19; 0-of-7 seed attempts complete) — the committee tool loop never terminates (an agentic orchestration gap, §7.5) — yet its 5 surviving rounds *look* spectacular. Both deficits are invisible to a returns-only view and are exactly what the D5 axis surfaces (Figure 8). The two have *different causes* — *gemini*’s is genuine schema-adherence, *qwen3*’s is scaffold-specific tool orchestration — which only the audit trail distinguishes.
- **F4** — Acuity is weak cohort-wide, with one exception. Under the v1.1 two-distractor noise test, D2 sits at $-0.07 \dots 0.08$ for eight of ten configurations (all but the haiku exception below and the qwen3 showcase seed) — not separable from a near-zero read at this sample size — and both qwen3 modes are negative (-0.06 autonomous mean, -0.79 on the structured showcase seed): their proposed trades track the distractors more than the alpha-bearing composite. The exception is *haiku · structured* at 0.44, the one configuration whose intent tracks the informative signals over both distractors amid the noisy 72-name universe — and, per F1, the same configuration whose narrative least explains its own trades. One caveat attaches: Acuity needs ≥ 3 simultaneous proposed tilts to score a round, and *haiku · structured*’s sparse-proposal style clears that bar in only a single round across its five seeds — so its 0.44 is a distinctive but thinly-sampled read, not a robust cohort-best.
- **F5** — Net of realistic costs, agents clear *some* of the passive suite, not the index cleanly. Reliable structured runs beat a median $\sim 6/8$ baselines (deepseek 6.8, haiku 6.0, gemini 6.0), but the beaten set is dominated by bonds / risk-parity / EW / 60-40; autonomous runs beat fewer (*qwen3* 2.2, *gemini* 3.0, *deepseek* 4.4). (*qwen3 · structured*’s 8/8 is on 5 rounds — discount it.) The instrument scores

against the *whole* suite, so “beats N/8” is honest rather than cherry-picked.

Methodology note. *max_decision_retries* was raised 1→3 mid-campaign after gemini’s residual schema-unreliability caused ~16% optimizer-fallback rounds; retries=3 cuts that to ~0 without masking the Reliability axis (D5 still records per-attempt failures). The live timeout/*decide* budgets were likewise widened for the reasoning-heavy models. All campaign runs use retries=3.

Cross-model analyses. The cross-model APM-CS radar (Figure 7), the single- and multi-module ablation Δ grid on the gemini anchor (§7.7, Figure 9), and the efficiency / cost-of- intelligence Pareto (Figure 14) are complete. Pairwise dominance / win-rate, the regime-stratified facet (risk-on / neutral / risk-off), and bootstrapped CIs / Kendall- τ rank-stability remain future passes (§8.8).

7.4 Radar Fingerprints (RQ2, continued)

The five-axis radar (Figure 7), split into structured (left) and autonomous (right) panels, visualizes the §7.3 finding that no configuration encloses all five axes. deepseek · structured is the most balanced enclosure (the APM-CS leader, 72.9) but does not top Coherence; the high-*Sharpe* configurations (qwen3 · structured, haiku · structured) are uneven, with strength on one or two axes and weakness on another (qwen3 on Reliability, haiku on Coherence). Capability is a *shape*, not a scalar. The two mode panels are descriptive fingerprints; D2 and D4 are mode-confounded and do not support direct cross-mode inference.

7.5 RQ3: Model-Specific Governance-Schema Adherence

Reliability is the axis that most cleanly separates the cohort — and it is invisible to returns. Three configurations carry reliability deficits — two are genuine governance-schema failures (gemini, deepseek-autonomous) and one is an agentic tool-orchestration failure (qwen3) — each surfaced only by the audit trail’s *parse_failures* / *decision_attempts* / *completion* fields:

- **qwen3-235b · structured** is the reliability floor — and the five-seed extension root-caused it as an *agentic tool-orchestration* failure, revising our earlier operational reading. Across **seven** seed attempts it never completes more than 10 of 26 rounds (2–10 per attempt; Reliability ≈ 0.2); two early hypotheses were eliminated in sequence — a 2048-token output cap truncated its structured JSON (fixed at 8192), and apparent provider throttling turned out to be a symptom, not the cause. The audit trail shows the mechanism: inside the structured committee’s bounded tool loop (10 turns/role) qwen3 *cannot terminate* — it re-invokes the same analysis tools with near-empty outputs (a typical failed seed logs ~12 tool-turn-budget exhaustions and ~16 null responses), the accumulated tool messages push the role context to its 32k cap, and the forced finalization returns null, killing the round. Decisively, the *same* model at the *same* context cap with the *same* tools completes **autonomous** runs cleanly (five complete seeds, zero nulls: the single-orchestrator loop is a simpler tool sequence it can drive), and the other four models clear the structured committee at the same cap with zero-to-one budget exhaustions. The

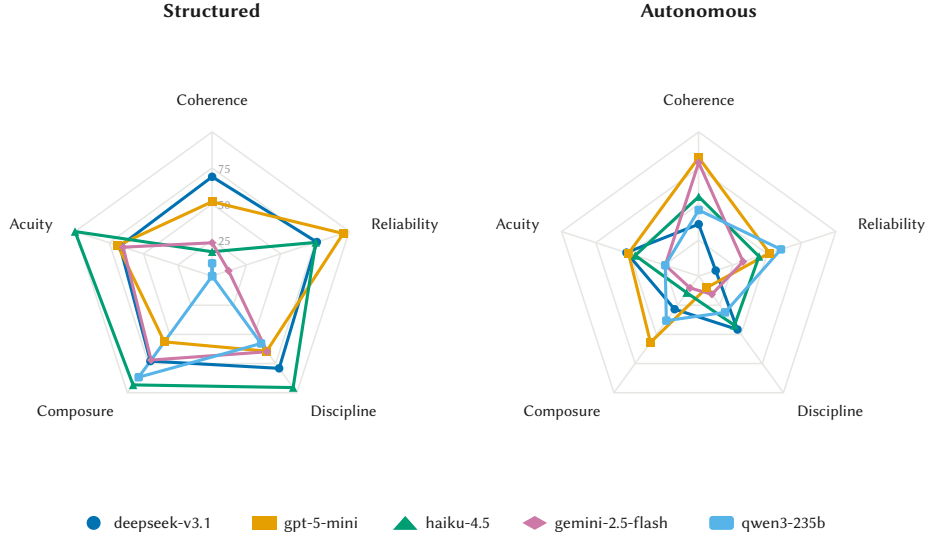


Figure 7. Five-axis diagnostic capability scorecard (within-cohort percentile per axis), all ten configurations split into structured (left) and autonomous (right) panels, one colour per model. No configuration encloses all five axes, and the composite leader (deepseek · structured) does not top Coherence. The panels are descriptive: D_2 and D_4 are mode-confounded and should not be interpreted as a clean structured-versus-autonomous effect.

scaffold, not the infrastructure, flips qwen3 from unusable to usable — a quantified scaffold-sensitivity data point. We therefore report qwen3 · structured only as a reliability/orchestration case study, not a capability row: its 9.82 Sharpe on 5 self-selected rounds is precisely the spurious number the scorecard exists to expose. (It runs fine in autonomous, 25/26.)

- **gemini-2.5-flash-lite · structured** posts a healthy +24.8% / Sharpe 2.31 (five-seed means, Table 5) yet Reliability 0.61 — residual parse-failures (~10/26 rounds on at least one attempt) persist across all its runs even with retries enabled. Retries recover the round, but the per-attempt schema failures are still recorded (as they must be) — strong outcome, measurable reliability deficit.
- **deepseek-chat-v3.1 · autonomous** inverts the pattern: excellent under the scaffold (structured Reliability 0.99) but 0.62 autonomous — the *same model* is reliable with the staged hand-off and substantially less reliable without it.

By contrast gpt-5-mini and claude-haiku-4.5 are schema-clean in both modes (Reliability ≈ 1.0). So governance-schema adherence is a real, model $\times$ mode-specific capability limit — the smaller models cannot reliably emit the structured decision — that a returns-only view misses entirely (gemini · structured beats ~6/8 baselines *while* failing the schema). This is precisely the kind of finding the benchmark exists to produce, now robust across repeated runs rather than a single pilot run.

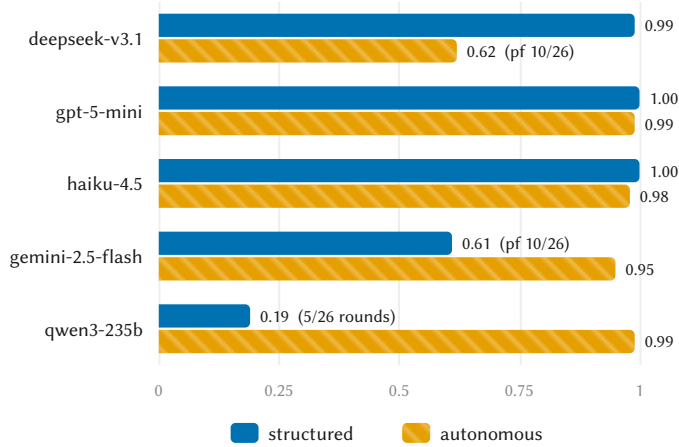


Figure 8. *Operational reliability (D5) by configuration, parse-fail-annotated. qwen3 · structured (2–10/26 $\rightarrow$ $\approx$ 0.2) and gemini · structured ($\approx$ 10 parse-fail rounds $\rightarrow$ 0.61) carry the cohort’s reliability deficits, both invisible to their returns; deepseek’s reliability flips with mode (structured 0.99 $\rightarrow$ autonomous 0.62). The “pf” labels mark mean parse-fail rounds, the mechanical driver of the D5 spread; qwen3’s incompleteness is the separate tool-loop non-termination failure (an agentic orchestration gap, §7.5), not a parse-fail.*

7.6 RQ4: Scaffold Shapes the Capability Profile

The skill mode reshapes the *capability* profile, and by markedly different amounts per model (§7.3-F2). The full cohort-relative APM-CS is higher for the four reliable structured backtest configurations, most dramatically for deepseek (72.9 vs 35.1), followed by haiku (71.0 vs 40.4), gpt-5-mini (67.4 vs 50.3), and gemini (47.4 vs 32.3). We treat these cross-mode composite contrasts as descriptive because D2 and D4 are mode-confounded; qwen3’s structured row (30.7) also rests on only five reliable rounds. The clearer axis-level result is that structure broadly lifts Composure (D3): structured configs hold one-way turnover flat in volatility (D3 $\approx$ 0.92–0.95), and the autonomous configs sit slightly lower (D3 $\approx$ 0.77–0.92), with gemini · autonomous the lowest at 0.77 — the one configuration with a positive volatility-overreaction slope (the others’ is $\approx$ 0). In this configuration, full autonomy coincides with greater volatility-sensitive turnover, a stability difference hidden by headline Sharpe. Acuity D2 and Discipline D4 are compared only within mode; their values are not used to identify a cross-mode effect.

On Coherence. The held-out judge exposes a weakness the agreement proxy alone would miss. Signal-action agreement is middling-to-fine cohort-wide ($\sim$ 0.53–0.66), but the reasoning-enabled minimax-m3 judge scores the allocations’ coherence *low*, so D1 sits in a 0.38–0.47 band for most configs — with claude-haiku · structured the outlier at 0.23: it posts the cohort’s *best reliable Sharpe* (3.21) and best Acuity (0.44) yet its allocations least follow its own stated analysis. The two Coherence components disagree — agents trade in broadly the right direction on average, but their narratives do not justify the

specific weights — and that disagreement is the diagnostic, visible only because Coherence pairs a mechanical agreement check with an independent reasoning-capable judge; either alone would mislead.

7.7 RQ5–RQ6: Ablation Grid (single- and multi-module)

The grid asks the returns question first — does removing a module move Sharpe beyond run-to-run noise? — and answers it in the negative for every module and every cluster (§7.7.1). It then turns to the axes that *do* register what removal costs (§7.7.2), and closes with the regime condition that bounds every returns-based reading (§7.7.3).

7.7.1 Single- and Multi-Module Knockouts: Returns Cannot Rank Module Value

Table 6. Ablation Δ Sharpe vs. the structured-gemini baseline (Sharpe 2.26 — the anchor seed’s run; the same configuration’s five-seed campaign mean is 2.31, Table 5; gemini anchor, bi-weekly, LOW tier). The six module-off variants are now run as repeated runs (the two cost-tier variants single-run); every run completed 26/26 rounds, chains verified, 0 failures. A ± 0.42 -Sharpe repeated-run noise band — $\pm 1\sigma$ of run-to-run Sharpe across the anchor configuration’s repeated runs — bounds separability: a variant is only separable if its whole run range clears the band — repeated runs move the composite by up to ~ 25 APM-CS points and Sharpe by up to ~ 2.3 across configurations, which is why each module-off Δ below is read against that band.

Variant	Δ Sharpe (mean)	run range	Reading
cost-high	−3.32	—	HIGH tier mis-calibrated (≈ 527 bps/rd $\rightarrow$ −75% cum, Sharpe −1.06); the one separable signal
cost-medium	−0.19	—	Δ cost tiny $\rightarrow$ decision noise
no-reflection	+0.09	[−0.16, +0.39]	within the noise band
no-news-sentiment	+0.12	[−0.34, +0.59]	within the noise band
no-advanced-quant	+0.16	[−0.06, +0.32]	within the noise band
no-cost-modeling	+0.37	[+0.36, +0.39]	tight, but still inside ± 0.42
no-memory	+0.52	[+0.24, +0.72]	straddles band on Sharpe — but drops Composure’s consistency leg (below)
no-world-macro	+0.66	[+0.33, +1.06]	largest mean, but range dips into the band $\rightarrow$ a favorable draw, not “macro helps”

(i) *Pipeline validated.* All module-off runs (six variants, repeated) plus the two cost-tier variants completed 26/26 with Δ computed against the verified baseline (Table 6).

(ii) *Cost realism (RQ6).* The HIGH (Almgren–Chriss [27]) tier is mis-calibrated: it charges ≈ 527 bps/round ($\sim 5.3\%$ /round $\rightarrow$ −75% cumulative, Sharpe −1.06), an implausible drag confirming the decision to keep LOW and flagging a cost-model recalibration task. MEDIUM moves cost negligibly, so its −0.19 Δ Sharpe is decision noise.

(iii) *Module importance is not rankable by Sharpe — even across repeated runs.* Every module-off Δ Sharpe is *positive* (means +0.09...+0.66) — removing a module never cleanly *hurts*, which is mechanistically implausible — and every run range straddles or sits inside the ± 0.42 band, so none is separable; the extra runs *narrowed* the estimates

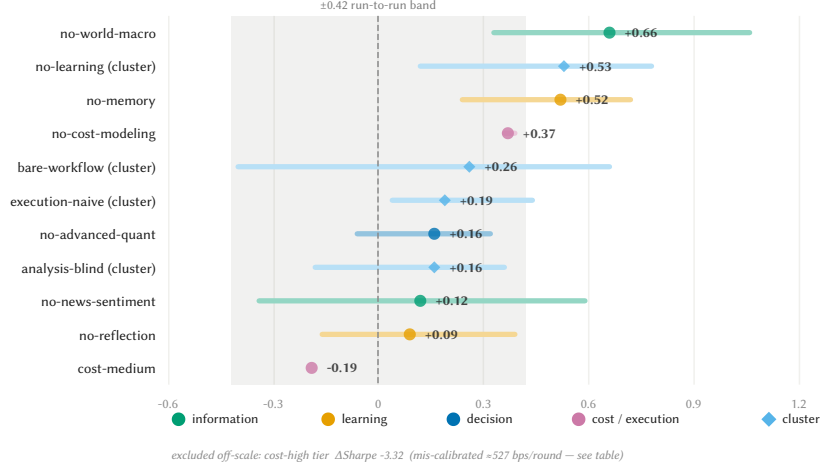


Figure 9. Ablation ΔSharpe (mean $\pm$ range across repeated runs) vs. the structured-gemini baseline, single-module and multi-module (hatched) knockouts. Only the mis-calibrated HIGH cost tier clears the ± 0.42 repeated-run noise band; neither single-module nor whole-cluster removal — down to the bare-workflow extreme — separates on returns.

without changing the verdict. *no-world-macro* (mean +0.66, range [+0.33, +1.06]) is the clearest *favorable draw*, not “macro helps.” Returns variance dominates the module signal in this fixed market path. Additional runs may detect smaller effects, but the present design does not separate them on returns.

(iv) *The capability scorecard catches what Sharpe cannot.* Removing the information stack leaves Sharpe in the noise yet collapses Acuity to ≈ -0.01 (the analysis-blind cluster, below): the agent has no informative signals left to attend to, a capability loss invisible to returns. Removing memory is subtler. It strips Composure’s *strategy-consistency* leg (the *consistency_score* only the memory module produces), so the axis falls back to its volatility-overreaction leg. Under the *old* multiplicative Composure this made the axis uncomputable, a construct defect the v2 revision fixes (§6.4); under v2 the dependency surfaces as the loss of the consistency leg rather than an undefined axis (and, because that leg sat below the calm leg in this benign window, the composite does not fall — the signal is the *absent leg*, read from the trail, not a lower number). So the ablation yields two robust signals — HIGH-tier mis-calibration (cost realism) and analysis-blind→Acuity (a capability dependency) — and a clean methodological result: module value is a multi-axis question that a returns-only ablation cannot answer, which is the entire reason the scorecard has five axes rather than one.

Multi-module (interaction) ablations. A single-module ablation cannot see *combined* contribution — modules may be individually redundant (the agent compensates when one input is gone) yet collectively essential. We therefore knock out whole capability clusters across repeated runs (12 more runs, all 26/26, same gemini anchor; Table 7):

Cluster knocked out (modules off)	Δ Sharpe (mean)	run range	CumRet	Acuity (D2)	Reading
analysis-blind (news + macro + advanced-quant)	+0.16	[−0.18, +0.36]	+27.1%	−0.01	within band on returns; Acuity collapses (no signal sources to attend to)
no-learning (memory + reflection)	+0.53	[+0.12, +0.78]	+29.6%	−0.01	within band; Composure loses its consistency leg (memory off)
execution-naive (cost-model + systematic-exec)	+0.19	[+0.04, +0.44]	+23.3%	0.12	within band; turnover unchanged (−0.18)
bare-workflow (only mandate + cost-aware execution)	+0.26	[−0.40, +0.66]	+27.3%	0.22	within band; the bare-workflow agent matches the full agent on returns

Table 7. Multi-module (cluster) ablation: Δ Sharpe vs. the structured-gemini baseline (Sharpe 2.26) when an entire capability cluster is removed (12 runs, all 26/26 rounds, same gemini anchor). Run range is the min–max across repeated runs; Reading names the capability axis the knockout most implicates. All four clusters stay inside the ± 0.42 repeated-run band on returns, yet the capability axes register what returns hide (analysis-blind $\rightarrow$ Acuity -0.01 , zero signal discrimination; v1.1 two-leg noise composite, §6.4).

(v) Combined removal does not separate on returns either, but the bare-workflow result is the striking case. None of the four cluster knockouts clears the ± 0.42 band, and every Δ Sharpe is again *positive* (+0.16...+0.53), so even removing an entire capability cluster never cleanly hurts. The extreme case is G4 bare-workflow: with *no* information modules, *no* learning, and even the quant-optimization engine removed — the LLM sets the weights itself under the mandate, with a risk-parity execution fallback — it is not separable from the fully-augmented six-module agent under the prespecified ± 0.42 repeated-run heuristic (Δ Sharpe +0.26, range [−0.40, +0.66]; +27.3% vs the baseline’s +24.9%). On *this* benchmark and period, the agentic scaffolding the system layers on contributes little to *Sharpe*. Yet — exactly as for the single-module grid — the capability axes register what returns hide: analysis-blind drives Acuity to ≈ -0.01 (the agent has no informative signals left to attend to), and both memory-removing clusters strip Composure’s strategy-consistency leg (the axis stays computable via its memory-free fallback, §6.4). (Bare-workflow’s *higher* Acuity point estimate — 0.22 vs the full agent’s 0.07 — is not a paradox: with the quant optimizer removed, the LLM’s proposal *is* the whole allocation move, so its signal-following shows up directly in the proposed deltas, whereas the full agent’s proposal is a tactical overlay on an optimizer baseline that has already absorbed much of the informative composite; and the estimate sits inside the config’s own large run spread of 0.04–0.44. We read it as a construction effect on what “proposed” measures, not superior signal attention.) The interaction test thus confirms, more forcefully, the thesis: *module value cannot be read off returns — not single-module, not cluster-level, not even against a stripped-to-the-mandate baseline — but the multi-axis scorecard does record what each removal costs.* (That a bare LLM matches the scaffolded agent on returns while differing in capability is itself a finding the benchmark exists to surface, and a caution against assuming added modules buy performance.)

7.7.2 Looking Past Sharpe: Risk, Cost, and Decision Behavior That no ablation moves Sharpe is not the end of the story but the start of it: a returns-only read would call all thirteen configurations equivalent when they in fact differ sharply on *risk*, *cost*, and — most of all — *decision behavior*. We diagnose each, because this is exactly the discrimination the instrument exists to provide.

(vi) *The full performance profile (Table 8) — three facts Sharpe hides.* (a) Genuine risk-adjusted skill, against the right benchmark: the full agent runs lower volatility (11.0% vs SPY’s 11.9%) and *half* the drawdown (−4.1% vs −8.9%) at the same Sharpe (2.26 vs 2.29) and $\approx 2\times$ the Calmar; its negative IR vs SPY (−0.04) is *deliberate de-risking* (giving up $\sim 2.4\%$ bull-market upside to halve the downside), an edge that inverts in a drawdown, not absence of skill. (b) That risk discipline is robust to ablation — every config but the mis-calibrated HIGH tier holds drawdown to −3.2... −4.7% and Calmar 5.8–8.7, with zero constraint-violations and style-drift across all thirteen — so it comes from the *constrained closed-loop* (caps + staged Risk-Officer/MAPC), not the optional modules. (c) The one module with a clean PnL footprint is cost-awareness: removing it raises realized cost +10% (the execution pair +18%); the cost *tier* matters where modules do not (HIGH is $122\times$ the LOW cost, vol 71%, DD −76%).

Table 8. *Multi-metric ablation profile (means over repeated runs; full = the structured-gemini baseline; SPY shown for the risk-adjusted comparison). Combos marked ⁺.*

Config	Sharpe	Sortino	Calmar	Vol%	MaxDD%Beat/8	IR vs SPY	Cost \$	Turn
<i>SPY (cap-wtd index)</i>	2.29	—	3.1	11.9	−8.9	—	—	—
full (all modules)	2.26	4.88	7.03	11.0	−4.1	5.7	−0.04	3 353
no-news-sentiment	2.38	5.31	5.91	11.0	−4.4	6.7	−0.01	3 338
no-advanced-quant	2.42	5.82	8.19	10.5	−3.3	6.3	−0.03	3 788
no-memory	2.78	5.33	7.00	9.0	−3.6	6.7	−0.04	3 657
analysis-blind ⁺	2.42	4.81	5.82	11.2	−4.7	7.0	−0.00	4 111
bare-workflow ⁺	2.52	5.48	7.16	10.9	−4.2	7.3	−0.00	3 983
cost-medium	2.07	4.85	4.57	13.3	−6.0	7.0	+0.01	4 787
cost-high	−1.06	−1.03	−0.99	70.9	−75.9	0.0	−0.33	411 154

(vii) *Behavioral diagnostics — detaching modules degrades the process, not the outcome (Fig. 10).* Stripping modules collapses decision scrutiny: the bare-workflow agent does half the analysis (research brief $0.51\times$), weighs half the signals ($0.48\times$), scouts no candidates and performs no reflection (both $0.00\times$), spends 40% less reasoning effort, yet proposes the largest trades (intended move $1.23\times$). On the *same* round (2025-11-03) the full agent records a mean-variance optimization with explicit weights, exclusions, binding constraints and a cost-management *lesson*, while the bare-workflow agent’s *entire* rationale is “Optimizer: risk_parity. Binding constraints: []. Accepted 0 adjustments” with no reflection (Appendix E) — premature construction, presumptuous intent, absent scrutiny. None of it shows in returns because the constrained construction contains it: concentration is flat (max-weight ~ 0.11 , HHI ~ 0.09) across all thirteen, so the rising presumptuousness surfaces in *proposed* moves the caps clip, never in *realized* positions. The guardrails convert degraded decisions into safe outcomes in a benign regime; the behavior and capability axes record what the constraints mask.

	analysis brief	signals weighed	candidates scouted	reflection	reasoning effort
bare-workflow	0.51×	0.48×	0.00×	0.00×	0.60×
analysis-blind	0.61×	0.59×	0.00×	1.11×	0.79×
¬quant	0.97×	0.70×	0.00×	1.07×	1.04×
¬reflect	1.02×	1.03×	1.00×	0.00×	0.82×
no-learning	1.03×	1.11×	1.00×	0.00×	0.83×
¬news	0.49×	0.78×	1.00×	1.15×	0.98×
¬cost	0.99×	0.90×	1.00×	0.90×	0.92×
¬macro	0.78×	0.97×	1.00×	1.10×	0.90×
¬memory	0.96×	0.93×	1.00×	1.16×	0.85×
exec-naive	0.93×	0.98×	1.00×	1.02×	1.03×

ratio to the full-module baseline (1.00× = unchanged; red = degraded, blue = elevated)

Figure 10. *Decision-quality proxies per ablation as a fraction of the full-module baseline (1.0 = unchanged; lower = more degraded). Stripping modules collapses analysis depth, signal breadth, candidate exploration, self-scrutiny and reasoning effort — the bare-workflow row is degraded on every axis — while returns are unaffected because the mandate’s constraints contain the degraded decisions (the full-performance profile, Table 8).*

(viii) *Module feasibility for a professional agent-investing platform.* Synthesising the returns / risk / cost / capability / behavioral evidence yields an actionable per-module verdict — the kind of guidance the benchmark is built to produce (Table 9), not a returns rank.

Table 9. *Per-module feasibility, grounded in the ablation evidence.*

Module	Measured effect of removal	Verdict (production platform)
<i>cost-modeling</i>	returns/risk-neutral; +10% realized friction ; HIGH tier −76% (mis-cal)	Keep — cheap, cost realism; <i>calibrate the tier</i> (HIGH is broken)
<i>systematic-execution</i>	neutral returns; turnover/cost ↑ when paired-off	Keep — execution discipline
<i>investment-target + constraints</i>	(not ablated alone) source of the −4% vs −9% drawdown edge ; contains overconfidence	Keep — essential ; the real risk-adjusted edge + the safeguard
<i>memory</i>	returns-neutral; zeroes lessons ; strips Composure’s strategy-consistency leg (D3, §6.4)	Keep — the only source of the strategy-consistency signal
<i>reflection</i>	returns-neutral; zeroes self-scrutiny/learning	Conditional — keep for auditability/improvement; deferrable for raw returns
<i>advanced-quant</i>	returns-neutral (risk-parity fallback suffices); zeroes candidate exploration, shallows analysis	Conditional — keep for breadth/optimization fidelity
<i>news-sentiment, world-macro, asset-class-data</i>	returns-neutral; feed Acuity (analysis-blind → −0.01) and analysis depth (¬news halves the brief)	Optional in benign regimes; keep for breadth/stress

The synthesis is the paper’s thesis in miniature: module value is real but lives off the

returns axis — in risk discipline (intrinsic to the constraints), in cost, and in decision *process* (depth, scrutiny, learning, measured intent). A returns leaderboard sees none of it and would prune the lot; the instrument prices each module on the axis it actually moves, and tells a platform builder *which* modules to keep, *why*, and *under what regime* the verdict flips.

7.7.3 The Regime Does Much of the Work (ix) *One benign regime masks both module value and model risk-taking.* Every number above is measured in one benign, rising market: over the campaign SPY returned +27% with only a −8.9% drawdown and a calm VIX (median 17; above 25 on just 7% of days). Three consequences follow, and each is a reason the ablation Δ are small. First, the market — not the modules — drives the returns: in a trending bull almost any reasonable long-only book posts a strong Sharpe and clears the defensive baselines, so removing a module *cannot* move returns much — there is little danger for the analytical or learning machinery to steer the agent away from. Second, the modules whose value is most regime-contingent are exactly the ones this period mutes: the information stack (*news-sentiment* / *world-macro* / *asset-class-data*) earns its keep at *turning points* — spotting deteriorating breadth or a regime shift — not while riding a trend; the learning loop (*memory* / *reflection*) pays off by *adapting across* regimes, and a single stable trend gives it little to adapt to; and the covariance-aware optimizer (*advanced-quant*) matters most when correlations spike in a sell-off, not when everything rises together. Their flat ablation Δ *here* is therefore weak evidence of irrelevance and strong evidence that this regime cannot test them. Third — and most consequential — a rising market masks risk-taking shortcomings the scorecard catches but returns do not. In a bull run the configurations that look best on raw return are those that take the *most* risk (full exposure, momentum-chasing, concentration), while the prudent de-risking the agents actually exhibit (half the index’s drawdown, Table 8) *costs* return and looks worse — so a returns leaderboard in this regime rewards exactly the recklessness a drawdown would punish and penalizes the risk management that would protect it. qwen3 · structured’s 5-round “Sharpe 9.82” (§7.3) is the extreme: an artifact a bull-market returns ranking flatters and the Reliability / risk axes expose. The benign window is thus not merely a power limit on the *ablations* (stress bucket $\approx$ 4–5 of 26 rounds, §8.7) but a structural reason the capability and risk-adjusted axes are necessary — they are what stays diagnostic when the regime makes returns uninformative, or worse, misleading.

7.8 Methodology note: reasoning needs output headroom

A test-time-compute model can look like an incapable portfolio manager when it is merely *truncated* — a tight output cap severs its chain-of-thought mid-stream. We therefore standardize on a generous output budget and report cap-hit, truncation, rate-limit, and parse-fail as first-class reliability telemetry. Without that telemetry a benchmark misattributes a budget or infrastructure artifact (qwen3’s incompleteness, §7.5) to a capability gap — exactly the misread a returns-only or single-configuration view invites.

7.9 RQ7: Net of Costs, Agents Clear the Defensive Baselines but Not the Index Cleanly

Over the full bi-weekly year, the reliable structured configurations clear a median ~ 6 of 8 passive baselines (deepseek 6.8, haiku 6.0, gemini 6.0; Figure 11) — a materially stronger showing than the single-quarter design pilot, where no agent beat SPY. But the beaten set is dominated by the suite’s defensive / diversified members (IEF bonds, risk-parity, 60/40, equal-weight); the cap-weighted equity index remains the hardest bar, and the autonomous configurations clear far fewer (qwen3-235b autonomous the lowest at 2.2/8 on seed means, gemini autonomous 3.0/8). The honest reading the instrument enforces — scoring every config against the *whole* suite, not a hand-picked benchmark — is that under realistic costs these agents reliably beat the defensive baselines but do not uniformly clear the cap-weighted index, and autonomy (autonomous mode) markedly weakens even that. (qwen3 · structured’s 8/8 is excluded as a 5-round artifact, §7.3.)

The index gap is a de-risking trade, not a deficit. “Does not beat the cap-weighted index” understates what the risk-adjusted read shows in Table 8: against SPY the agents run lower volatility and roughly *half* the maximum drawdown (-4.1% vs -8.9%) at the same Sharpe and $\approx 2\times$ the Calmar. The shortfall in *absolute* return is the price of deliberate diversification away from the index’s mega-cap concentration — a trade that *loses* a little in a concentrated bull run and *wins* decisively in a drawdown. So the honest statement is not “no skill vs the index” but “a risk-managed profile the index does not have”: the agents convert a passive long-only mandate into a lower-drawdown, comparable-Sharpe strategy, and — per the ablation diagnostics — this risk discipline is intrinsic to the constrained construction and survives even the bare-workflow stripping. The Composure axis (§7.6) is the in-instrument measure of exactly this behavior; a returns leaderboard, fixated on the -2.4% vs SPY, would miss it.

7.10 Live-Track Decision-Level Validation (Completed Four-Week Campaign, 18 Valid Days)

The live track ran as a four-week daily-rebalance campaign on a shared broker paper account, now complete: **eighteen valid NYSE days** (2026-06-15 $\rightarrow$ 2026-07-10): 6 newer “thinking” models (minimax-m3, deepseek-v4-pro, qwen3.7-plus, gpt-5.4-mini, gemini-3.5-flash, claude-sonnet-4.6) $\times$ {structured, autonomous} = 12 concurrent virtual-ledger sessions, each a \$1M slice at medium reasoning (2026-06-19, Juneteenth, is dropped via a published exclusion manifest enforced by an exchange-calendar gate; the July-3 market holiday is skipped by the same gate). Each ledger opens from an equal-weight core book at the prior close, only the daily delta reaches the broker, and an end-of-day two-layer flatten returns the shared account to flat. Because the live cohort is a *newer generation* than the backtest cohort and realized P&L is a sequence of one-day deltas, we treat live returns as indicative-only and evaluate at the *decision level*: does the instrument produce coherent, interpretable capability diagnostics on data the models never saw? Eighteen rounds — up from a four-round week-1 pilot, a nine-round two-week read, and a fourteen-round interim draft — let the live axes resolve rather than merely sketch, and expose how the shortest-horizon axes keep moving into the campaign’s final week (below).

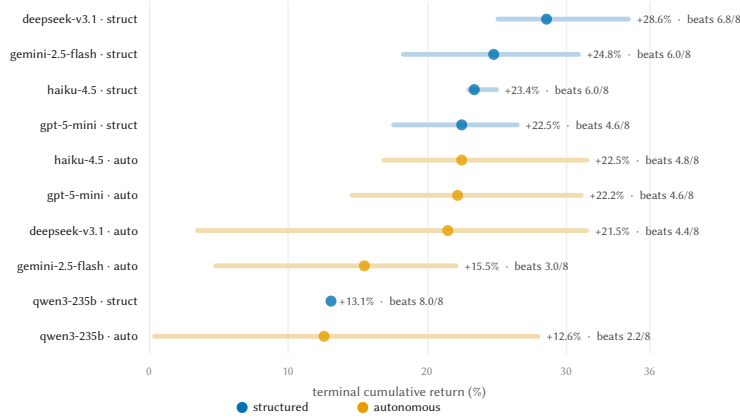


Figure 11. *Terminal cumulative return by configuration over the 26 bi-weekly rounds (2025-06 → 2026-06); whiskers span the range across repeated runs, and each label gives the number of the 8 passive baselines the configuration beats. Most structured configs sit above their autonomous counterparts on terminal return and clear the defensive baselines ($\approx 6/8$), while autonomous configs trail (qwen3-235b autonomous the lowest at 2.2/8 on seed means).*

Campaign integrity ledger. Three operational events are disclosed and marked in the data rather than silently absorbed. (i) *Mid-campaign remediation boundary* (the 2026-07-02 session — the 13th of the 18 valid trading days; ledger round 14, the ledger count including the excluded holiday round). A prompt-layer and failure-handling audit was remediated mid-campaign: the autonomous path gained output-schema coercion and fail-loud decision retries (previously a failed decision silently degraded into a hold, §7.11), the Risk Officer’s prompt gained the mandate’s actual constraint numbers, and the memory loop was closed with bounded lesson/history feedback (previously write-only at the prompt level). Rounds 1–13 ran on the earlier prompts, so prompt-sensitive quantities are read with the boundary marked: the pooled eighteen-round scorecard is the primary read, and the one metric the boundary visibly moves — the autonomous hold rate — is reported split in §7.11, where the boundary doubles as an intervention test of the hold mechanisms. (ii) *Broker-gateway outage* (2026-07-06). A gateway failure broke contract-id resolution at EXECUTE for the six autonomous sessions *after* their decisions were recorded; EXECUTE alone was re-run the same day against the unchanged recorded targets (regular-session fills; each round’s hash chain re-verified). Decision-level scoring is unaffected by construction. (iii) *Per-model reasoning deviation* (final round, 2026-07-10). qwen3.7-plus’s structured PM entered a non-terminating reasoning/tool loop at the standard medium effort (single calls of 200–770 s or timing out, reproducible in isolation), so its final round ran in both modes with reasoning disabled while the other ten sessions stayed at medium — an echo, in a second qwen generation, of the qwen3-235b structured tool-loop non-termination of §7.5. All twelve ledgers verify end-to-end (19 recorded rounds each, hash chains PASS).

Decision-level scorecard (live). Computing the five axes over the eighteen valid rounds

(Coherence via the held-out GLM-5.2 judge, the live primary under the judge $\neq$ cohort protocol §6.4.1; Table 10) separates the live cohort along the same dimensions as backtest — and, as in backtest, no configuration dominates all five axes. claude-sonnet-4.6 leads (autonomous APM-CS 83.6), with qwen3.7-plus (69.1) next and deepseek-v4-pro and gpt-5.4-mini tied at 63.6; deepseek-v4-pro autonomous carries the cohort’s best judge leg (0.67). The scaffold split in the composite is uniform: **every model’s five-axis composite — whose Coherence axis carries the held-out judge’s leg — is higher autonomous than structured** (sonnet 83.6 vs 52.7; qwen3.7 69.1 vs 40.0; deepseek 63.6 vs 50.9; gpt 63.6 vs 36.4; gemini 43.6 vs 25.5; minimax 41.8 vs 29.1) — a split that held at nine, fourteen, and eighteen rounds, and that persists on the mode-comparable D1/D3/D5 sensitivity (Appendix F.4). The diagnostically richest rows are the floors: minimax-m3 remains the Reliability floor (0.49 autonomous / 0.56 structured; parse failures in both modes, 9 and 7 of 18 rounds), and Composure settles into a near-bimodal split. On its floor sit gemini (0.44/0.46) and minimax (0.45/0.47) in both modes, with gpt-5.4-mini · structured beside them (0.49); sonnet, qwen3.7 and deepseek hold ≈ 0.93 – 0.95 in both modes, and gpt’s autonomous side sits between (0.69). The sharpest horizon effect is gemini-3.5-flash · structured: mid-pack at nine days, it fell to the cohort floor at fourteen and stayed there through the finish (APM-CS 25.5) as its Composure read collapsed — while deepseek’s and gpt’s autonomous Composure rose through the final week: the shortest-horizon axes keep moving to the end of even a four-week record. The autonomous hold behavior behind minimax’s reliability hit — and gemini-3.5-flash’s distinct hold mechanism, both eliminated after the round-14 remediation — is dissected in §7.11.

Table 10. *Live decision-level scorecard (18 valid NYSE days, 2026-06-15 $\rightarrow$ 07-10, the completed campaign; Coherence = $\frac{1}{2} \cdot \text{signal-action agreement} + \frac{1}{2} \cdot \text{held-out GLM-5.2 judge, the live primary per §6.4.1}$; D2 uses the v1.1 two-leg noise composite; APM-CS = mean cohort percentile across the five axes). Returns are indicative-only and omitted. Sorted by APM-CS; per-axis leader bold.*

Model · mode	D1 Coher.	D2 Acuity	D3 Compos.	D4 Discip.	D5 Reliab.	APM-CS
claude-sonnet-4.6 · auto	0.60	0.30	0.94	0.80	1.00	83.6
qwen3.7-plus · auto	0.59	0.26	0.94	0.75	1.00	69.1
deepseek-v4-pro · auto	0.62	0.14	0.95	0.62	1.00	63.6
gpt-5.4-mini · auto	0.56	0.27	0.69	0.83	0.99	63.6
claude-sonnet-4.6 · stru	0.52	0.14	0.93	0.95	0.82	52.7
deepseek-v4-pro · stru	0.50	0.09	0.95	0.68	1.00	50.9
gemini-3.5-flash · auto	0.51	0.29	0.44	0.65	1.00	43.6
minimax-m3 · auto	0.56	0.25	0.45	0.91	0.49	41.8
qwen3.7-plus · stru	0.41	−0.10	0.95	0.95	0.58	40.0
gpt-5.4-mini · stru	0.48	0.10	0.49	0.78	0.97	36.4
minimax-m3 · stru	0.41	0.00	0.47	0.97	0.56	29.1
gemini-3.5-flash · stru	0.42	0.18	0.46	0.58	0.94	25.5

The cross-track invariant (headline live result). The mechanical signal-action *agreement* and the held-out reasoning *judge* — the two components of Coherence — diverge systematically on both tracks: **+0.30 on the backtest** (agreement 0.58 / minimax-m3 judge 0.28, five-seed cohort) and **+0.23 on the live track** (agreement 0.63 / GLM-5.2 judge 0.40 over the completed eighteen rounds; Fig. 12). Because

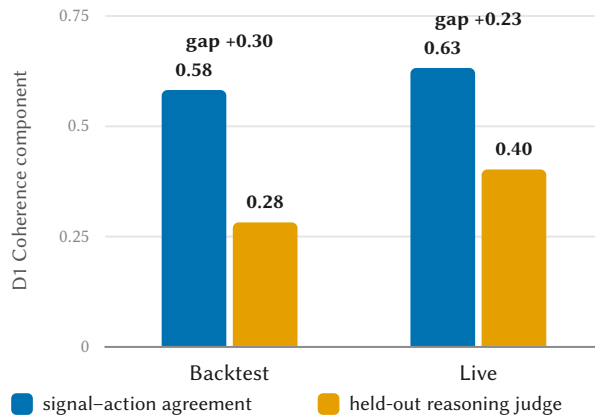


Figure 12. *The agreement↔judge coherence gap is +0.30 (backtest, minimax-m3 judge) and +0.23 (live, GLM-5.2 judge, eighteen valid rounds — the completed campaign; protocol-clean within-judge comparisons stay in the same +0.23...+0.30 band) — across two cohorts, two horizons, and two held-out judges the instrument measures the same decision property on data the models never saw.*

each track has a different primary (§6.4.1), the cleanest read is *within* judge, and both judges independently show the same structure: GLM-5.2 — the only judge with protocol-clean reads on both tracks — sees +0.27 (backtest corroboration) vs +0.23 (live), and minimax-m3 reads +0.30 on the backtest, its one protocol-clean read. Across the two judges, two cohorts, two periods, and two cadences the protocol-clean gaps span a narrow +0.23...+0.30 band (GLM uniformly ~0.05 more lenient). One superseded read sits outside it: minimax-m3’s v1.0 read of the live track — taken before the per-track hold-out was enforced, so it scored a cohort containing itself — was +0.34 and is excluded from the primary band; its direction argues against self-preference inflation, as it scored its own rounds among the cohort’s lowest (Appendix C). What persists is the same systematic divergence between *agreeing with signals mechanically* and *cohering with one’s own stated reasoning*, measured on genuinely unseen data (not a claim that any single model generalizes — §8.7). Newer live models lift both components in step (deepseek’s judge leg rises 0.37 → 0.67 from backtest-era v3.1 to live v4-pro under the same GLM read, with agreement rising too), so the *difference* persists even as both levels lift — and the gap has been pinned at the same magnitude through every horizon extension (+0.25 at nine days, +0.23 at fourteen, +0.23 at the completed eighteen), still far from closing.

What the gap measures, and why coherence scores low. The two halves of Coherence ask different-difficulty questions. *Agreement* checks only the direction of each proposed trade against the sign of that name’s informative composite — a low bar a coin-flip clears at 0.5 and modest skill lifts to ~0.6. The *judge* asks whether the specific weights are a coherent *consequence* of the agent’s own stated analysis — and the cohort fails it: across the year-long backtest campaign the held-out judge’s mean score is 2.12 of 5 (mode 2; normalized 0.28), and the newer live cohort reads higher under GLM-5.2

(0.40 normalized) while still trailing its own agreement leg. We treat this **two-leg divergence** descriptively: the components are bounded to the same interval but have different baselines and difficulty, so their numerical difference is not a calibrated effect size. It nevertheless separates two competencies — *aligning with external signals* and *keeping one’s allocation consistent with one’s own narrative* — and its stability is the empirical result. Reading the cached judge rationales, the low scores are concrete and checkable, recurring in five patterns: (i) direct contradiction (“analysis says TSLA should be *reduced* but its weight change is +1.7%”); (ii) equal-weight collapse that ignores the differentiation the analysis emphasized (“near-equal-weight 7.1–7.3% across all names ... CAT +3.74 vs LIN −3.59 receive essentially identical weights, incoherent with a signal-driven mean-variance analysis”); (iii) name mismatch (the thesis names AVGO/GS/MS/NEM; the book holds PG/AMT/NEE/AMZN); (iv) weights anti-correlated with the cited signals (overweighting the most bearish name at 12% while the strongest bullish name gets 0.2%); and (v) no thesis at all (an autonomous→MAPC placeholder with “no thesis for the specific name selection”). This is the *stating-versus-doing* deficit made concrete.

Where the incoherence enters. Part of it is the agent’s narrative not matching even its own intent; part is the *machinery*. The judge scores the post-construction target, which the quant optimizer and the constrained projection have already reshaped — so the equal-weight collapse (the optimizer flattening a tactical tilt) and the cap-hits (the projection clipping) are partly system effects, not pure reasoning failures. The instrument records *proposed* (pre-MAPC intent) separately from *projected* (post-machinery) precisely so the two can be attributed (§3.7); the extreme tell is the bare-workflow agent (§7.7) whose entire rationale is “*Optimizer: risk parity ... Accepted 0 adjustments*” — there is no thesis for the allocation to cohere with. Read at the level of the agent-in-the-system, then, the gap measures a real reasoning–action consistency deficit: a competent reader, handed the agent’s analysis, would not derive the book it produced.

What the cross-track agreement does and does not establish. The divergence persists on genuinely post-cutoff live data, making a period-specific or leakage-only explanation less plausible without establishing same-model transfer. Under the same GLM judge, deepseek’s judge leg rises $0.37 \rightarrow 0.67$ from backtest-era v3.1 to live v4-pro and agreement rises in step, leaving a similar difference. This pattern is consistent with, but does not test, the declarative-to-procedural interpretation motivated by ACT* (§2.4). The per-track gaps are *similar, not identical* (+0.299 backtest vs +0.230 live, under different primaries; within judge, GLM reads +0.27/+0.23 and minimax +0.30 on its protocol-clean backtest read) — consistent with one systematic deficit measured at judge-dependent levels rather than a proof of equality; and because both tracks share the metric, the match is instrument-consistency, a statement about the apparatus measuring stably. The strongest robustness check is therefore the second judge (below), which rules out a different failure mode — one judge’s idiosyncrasy — than the cross-track test does.

What the cross-track does *not* claim. The live and backtest cohorts are different model *versions*, so this is instrument-consistency, not same-model generalization; family-level traits reflect model *and* condition. The extending horizon also kept revising the shortest-baseline axes — a caution the campaign documents repeatedly: week 2 lifted gemini-3.5-flash’s Composure from the week-1 floor to the cohort top, weeks 3–4 pulled it

back to the floor where it stayed through the finish (0.44 autonomous / 0.46 structured at eighteen rounds), and the final week substantially lifted deepseek’s and gpt-5.4-mini’s autonomous Composure reads, while gpt-5.4-mini’s week-1 scaffold edge stayed reversed. Coherence, Reliability, and Discipline have been stable across all four reads; Composure and Acuity remain the shortest-horizon axes and are read with proportionate caution — against over-reading any short live record, our own earlier reads included.

Robustness to the judge. Because the coherence gap is the instrument’s strongest claim, the two per-track primaries are calibrated against each other on the backtest rounds both have scored (full protocol in §6.4.1). Over 633 shared rounds the judges agree closely (Spearman $\rho = 0.65$, Pearson $r = 0.66$, 97% within one point on the 1–5 rubric; GLM runs ~ 0.2 rubric-points more lenient, mean 2.32 vs 2.12), and the gap survives a full judge swap on the backtest — +0.27 under GLM vs +0.30 under minimax-m3 — so the divergence is a property of the *allocations*, not of one particular judge.

7.11 Behavioral Interpretability: Findings

Because every round stores the agent’s stated analysis, rationale, reflection, and parse status, the trail lets us read *behavior*, not just scores — and the most instructive live behavior is an autonomous-mode convergence to hold (no rebalance) that, examined closely, decomposes into two distinct, model-specific mechanisms. The structured scaffold prevents both.

Two roads to the same hold (Fig. 13). Across the eighteen valid live days, exactly two of six models ever produce a null-strategy hold, and only in autonomous mode: gemini-3.5-flash holds 9 of 18 rounds and minimax-m3 8 of 18, while the other four autonomous models — and all six in structured mode — hold 0%. The two roads are distinct: - *gemini-3.5-flash* — *tool-turn-budget exhaustion*. Its reflection carries a complete mean-variance thesis (ten high-momentum names, a risk-on regime read), yet the committed round records only a truncation marker: the autonomous agent spends its tool-turn budget exploring and never serializes a final allocation, so an empty-weights fallback holds the book. It *reasoned* but did not *deliver in budget*. - *minimax-m3* — *governance-schema-adherence failure*. The round is rejected for emitting schema-invalid output — an empty value where the output schema required a list — and with no valid decision it falls back to hold.

Intermittent, independent, recoverable — not a one-way collapse. The first live week reads like a clean “day-1 trade, then lock-up”: both models traded day 1 and then held days 2–4 in lockstep, which a short pilot would report as a deterministic day-2 collapse. The second week overturns that reading. The holds are intermittent (both models trade on several later days — gemini on 06-23/06-24, minimax on 06-23/06-25/06-26 — so the failure is *recoverable*, not absorbing), and they are independent: in week 2 the two models hold on *different* days (06-24 minimax holds while gemini trades; 06-25/06-26 gemini holds while minimax trades), with no synchrony. The diagnostically correct summary is therefore a *per-model autonomous hold-rate* (gemini 0.50, minimax 0.44 over the completed campaign) driven by a stochastic, model-specific fragility the autonomous scaffold exposes — *not* a shared, monotone “models stop trading after day 1.” This

	Structured	Autonomous
minimax-m3	0%	44% schema fail · 06-16
deepseek-v4-pro	0%	0%
qwen3.7-plus	0%	0%
gpt-5.4-mini	0%	0%
gemini-3.5-flash	0%	50% tool-turn · 06-16
sonnet-4.6	0%	0%

Figure 13. *Live hold-round rate by model × mode over the eighteen valid days. Only two models ever hold, and only in autonomous mode — gemini-3.5-flash (0.50, tool-turn truncation) and minimax-m3 (0.44, schema-adherence failure) — via different mechanisms; the holds are intermittent, recoverable, and confined to the pre-remediation segment (all 17 hold rounds precede the round-14 scaffold fix, after which both models trade every day, §7.11). Structured mode = 0% for all.*

is exactly the kind of claim a longer live horizon is needed to get right, and a caution against over-reading any few-round live result, our own week-1 pilot included.

The remediation boundary turns the diagnosis into an intervention test. Both mechanisms were root-caused mid-campaign from the audit trail exactly as described above, and the round-14 remediation (§7.10) deployed the corresponding scaffold-level counter-measures: output-schema coercion for the list-type failure, and fail-loud decision retries — with the previous failure named in the retry prompt — in place of the silent empty-weights fallback. The observed change is sharp: every one of the 17 hold rounds precedes the boundary (gemini 9 of 12 pre-boundary valid days, minimax 8 of 12), and neither model holds in the six valid days from the boundary session (2026-07-02) through the campaign end (0 of 12 model-days) — with both models untouched. Re-running an individual failed round had reproduced the failure under the earlier scaffold. Their disappearance after the targeted, prospectively deployed remediation is therefore mechanism-consistent intervention evidence that the holds arose from scaffold-level failure handling interacting with model-specific output fragility, rather than from strategic choices. It is not a general efficacy estimate: the intervention is a bundled scaffold change observed for two models over six post-boundary days without a parallel control. Historical rounds stay as recorded (the live series is path-dependent), flagged by the scorer and excluded from the consistency axis.

Scaffold magnifies vs. constrains (the structured–autonomous thread). The same minimax-m3 in *structured* mode still logs parse failures yet keeps trading: the staged pipeline (Risk-Officer review, MAPC reconciliation) contains a malformed sub-output and recovers,

where the single-shot autonomous path lets the same failure collapse the whole round. This is a concrete instance of a recurring theme — structure converts a model’s raw fragility into graceful degradation, at the cost of the autonomy that, for the more capable models, yields the cohort’s *best* decisions (claude-sonnet-4.6 and qwen3.7-plus score markedly higher autonomous than structured — 83.6 vs 52.7 and 69.1 vs 40.0 — and *every* live model’s five-axis composite is higher autonomous, Table 10). The composite and the containment story pull apart for the fragile pair: minimax-m3’s autonomous APM-CS still exceeds its structured one (41.8 vs 29.1) because its structured committee legs are weaker still — the scaffold’s value there shows in *which* failures occur (it keeps trading, gracefully degraded), not in the composite. Whether an agent is best deployed as a disciplined expert-human proxy (structured) or given autonomy to exceed human heuristics is therefore model-dependent, not settled — a question we return to in §8.

Competence, read fairly. The trail is not only a catalogue of failure modes. claude-sonnet-4.6’s day-1 rationale is crisp and executable: it reads a risk-on regime (VIX ~18), flags a negative momentum loading in its factor model and rotates to correct it, and exits its worst-momentum, highest-volatility names — a defensible thesis that the allocation then follows, and the scorecard rewards it (Coherence 0.60, APM-CS 83.6 — the live leader). The same field that exposes minimax’s schema failure credits claude’s discipline; that even-handedness is the point of reading the trail.

Every quote above is pinned to a hash-verified *round_id* (anecdote bank, Appendix E).

Each behavioral lens of §6.5 maps back to a capability axis (stated reasoning → Coherence; information foraging → Acuity; struggle signatures → Reliability), so the qualitative read *explains, in the model’s own words and actions, how* the quantitative deficits arise. A fuller per-model behavioral table (clip-rate, RO-override rate, mean tool-turns, lessons/round, retry-rate) is a natural extension of this trail; here we report the most diagnostic slice — the hold-mechanism split (Figure 13) and the anecdote bank (Appendix E) — and leave the exhaustive tabulation to future work.

7.12 Summary

On clean, chain-verified campaign data (5 models × 2 modes; five seeds per reliable configuration; a 13-configuration ablation grid; held-out judge) the instrument (i) discriminates decision quality where Sharpe conflates it — the capability leader (deepseek · structured, APM-CS 72.9) is not the Sharpe “winner” (qwen3 · structured 9.82, exposed as a 5/26-round artifact) (F1); (ii) explains outcomes through measurable deficits — qwen3’s structured tool-loop non-termination and gemini’s persistent parse-fails (Reliability), the autonomous-mode Composure churn, and near-zero cohort Acuity outside haiku (F3–F4); (iii) shows scaffold reshapes capability by model-specific amounts (deepseek –38 APM-CS from structured to autonomous) (F2); and (iv) surfaces benchmark limitations the campaign was built to expose — even repeated-run ablations cannot rank module value *by returns* (though the capability axes catch analysis-blind→Acuity), and the HIGH cost tier is mis-calibrated (§7.7). No configuration dominates all five axes, so by the instrument’s own standard the cohort is not cleanly rankable — the intended demonstration that diagnosis must precede ranking. The live track (§7.10–§7.11) adds

the out-of-distribution leg of the same argument: the agreement $\leftrightarrow$ judge coherence gap reproduces on genuinely unseen data at similar magnitude (+0.30 backtest / +0.23 live over the completed eighteen-round campaign), and the autonomous-only hold behavior resolves into two distinct, model-specific mechanisms whose disappearance after a targeted mid-campaign scaffold fix (zero holds in the final six valid days) provides mechanism-consistent intervention evidence — closing the validation arc from backtest, through ablation, to live execution.

8. Discussion

8.1 Diagnosis Before Ranking

The campaign makes the central argument concretely: the capability leader (deepseek · structured, APM-CS 72.9) and the nominal Sharpe “winner” (qwen3 · structured, 9.82) are different configurations — and the latter is a 5-of-26-round artifact (Reliability 0.19) the scorecard exposes rather than crowns. A leaderboard would report a spurious number and stop; the scorecard instead returns a *profile* for each — qwen3 attends to the distractors over the alpha signal (negative Acuity) and cannot *terminate* the structured committee’s tool loop (an agentic orchestration gap, root-caused across seven failed seed attempts, §7.5); haiku · structured posts the best *reliable* Sharpe (3.21) and the cohort’s best Acuity (0.44) yet the lowest Coherence (0.23); deepseek leads on balance but is only mid-pack on Sharpe; and the *same* model’s profile shifts sharply with scaffold (deepseek 72.9 structured $\rightarrow$ 35.1 autonomous). These are *research-actionable* statements (where to intervene), not a rank. They also explain *why* we decline to rank: no configuration dominates all five axes, and a credible cross-regime ranking would need an order of magnitude more models, regimes, and repeated runs — out of scope by design, and exactly the larger evidence base the instrument is meant to support over time.

8.2 Capability Axes as General Agent Diagnostics

The axis $\rightarrow$ construct mapping. Each axis is deliberately a finance instantiation of a general capability, which lets trading findings speak to model behavior at large. Coherence is *systems-level reasoning vs. greedy local moves* — the gap between an agent’s stated analysis and its action; the judge/agreement divergence (§7.10) is a finance reading of “the model can name the right consideration but does not let it govern the output.” Acuity is *salience attention under polluted context* — the negative-Acuity autonomous run is a finance reading of distractibility amid abundant, partly misleading data. Composure is *robustness to perturbation vs. overcorrection* — gemini-autonomous’s vol-driven churn (§7.6) is reactive overcorrection. Discipline is *systematic alignment to a standing mandate vs. turn-by-turn instruction-following* — self-regulation without a hard guardrail. Reliability is *protocol/format adherence and operational robustness* — completing a structured tool-use task in valid form, every time.

The ACT* reading of the coherence gap. Read through Anderson’s ACT* lens [25], the coherence deficit is declarative $\rightarrow$ procedural: the agent holds the *declarative* knowledge (it states the correct consideration — “VIX is elevated, reduce momentum exposure”) but lacks the compiled *procedural* production rule that would convert that knowledge

into the specific weights — exactly the failure ACT* predicts for knowledge acquired from text without compilation through practice. On this reading the coherence gap, reproducing across backtest and live at similar magnitude (+0.30 / +0.23; protocol-clean within-judge band +0.23...+0.30, §7.10), is — as the most parsimonious interpretation we offer, not a tested causal claim — consistent with the deficit rather than measurement noise: the declarative store transfers cleanly to new data (the agent still names the right factors) while the missing procedural compilation transfers too (the actions still do not follow), so the gap reproduces at the same magnitude on data the models never saw. The reflect-and-learn loop (the ALE stage and three-tier memory, §3.6) is then the mechanism by which a benchmark could, in principle, supply the practice ACT* says is required — making the memory ablation (§7.7) a direct test of whether in-context experience can begin to compile the missing rules.

A portable lens. Read this way, the same five questions can be asked of any agentic system operating a complex, cost-bearing, partially observable process.

Parallels in the benchmark literature. Three of CLQT’s findings rhyme with results obtained by very different instruments, which is mild evidence they are properties of the *agents*, not of our setup. First, the coherence gap echoes Profit Mirage [8]: where they show apparent alpha *collapses once leakage is removed*, we show apparent skill *fails to cohere with stated reasoning* — both are cases of a headline number evaporating under a closer look, and both argue for measuring the process rather than trusting the outcome. Second, the capability $\neq$ outcome result (the Sharpe “winner” is a 5-round artifact, §7.3) is the portfolio-management instance of procedure-aware evaluation’s “corrupt success” [12] — task success reached through an unsound process — which CLQT surfaces because the audit trail scores the process, not just the PnL. Third, the repeated-run noise floor (§7.7) is a concrete instance of the statistical-power cautions that agent-evaluation surveys raise in the abstract: a single-axis, single-run score is not just incomplete but *unstable*, which is why CLQT reports a profile, bounds module-importance with a repeated-run protocol, and treats reliability as a first-class axis.

8.3 Scaffolding $\times$ Capability, and the Cost of Intelligence

The structured $\leftrightarrow$ autonomous contrast is the clearest interaction in the campaign: structure lifts the capability profile broadly (Composure, Discipline), most dramatically for deepseek (APM-CS +38). But it also *demand*s a governance schema that stresses the smaller models in different ways (gemini’s genuine parse-fails; qwen3’s tool-loop non-termination, §7.5). Structure thus *helps and hurts the same model on different axes*, and for deepseek its reliability reverses with mode (reliable structured, less reliable autonomous). That is a more useful statement than “scaffolding helps weaker models,” and one only a multi-axis read can make. The efficiency ledger sharpens it: gemini $\cdot$ structured is by far the heaviest reasoner (~280k reasoning tokens/run despite thinking being pinned off), its compute consumed by the retry loop around its schema failures. “Cost of intelligence” here is dominated by *failed* reasoning, not productive reasoning (Figure 14).

Which investment schema — structured, autonomous, or hybrid? The live month

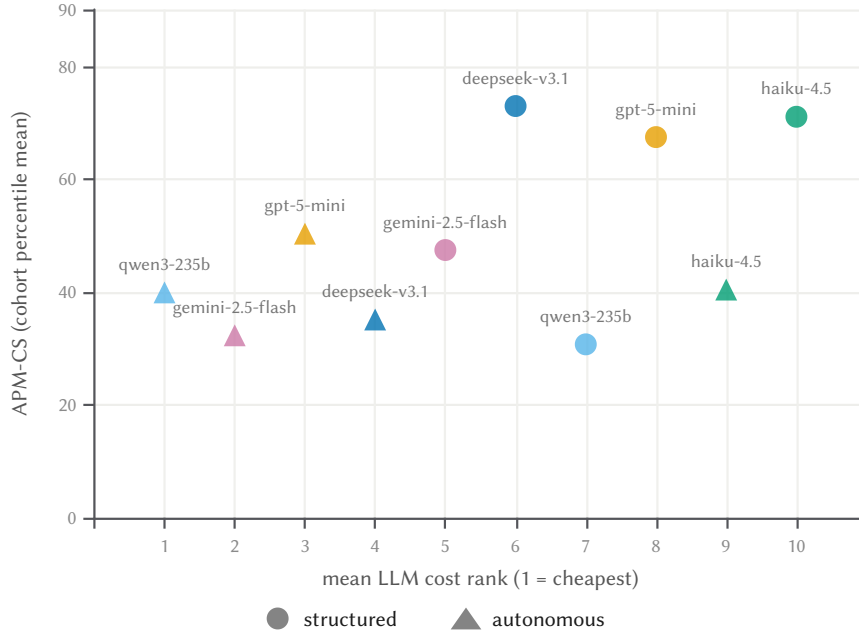


Figure 14. *Capability (APM-CS) vs. average LLM cost rank (1 = cheapest); circles = structured, triangles = autonomous. deepseek · structured delivers the highest capability (72.9) among the cheaper configurations, while haiku · structured reaches comparable capability (71.0) at several times the cost, and gemini · structured is the heaviest reasoner yet only mid-capability — its compute largely consumed by the parse-fail retry loop. Capability does not track spend.*

sharpens the structured↔autonomous contrast into a design choice. Structure acts as a *containment* layer: it converts a model’s raw fragility into graceful degradation (minimax-m3 keeps trading in structured mode but collapses to holds when autonomous, §7.11) and lifts Composure and Discipline broadly. Autonomy removes that scaffolding — which *destabilizes* the models that cannot self-govern (gemini-3.5-flash and minimax-m3 collapse to intermittent holds; minimax’s Reliability falls to 0.49) even though the held-out judge scores every live model’s composite higher autonomous than structured (Table 10): the gains are largest and cleanest for the capable self-governors (claude-sonnet-4.6 83.6, qwen3.7-plus 69.1, both at full Reliability), while gemini’s and minimax’s smaller autonomous edges ride on the rounds their hold-collapses spare. The implication is that the right schema is model-dependent, and the interesting frontier is neither pole but a hybrid — let an agent operate autonomously while a lightweight governance layer (schema validation, mandate reconciliation, a held-out coherence check) catches exactly the failure modes structure would have prevented, keeping the upside of autonomy without its brittleness. The campaign closed with a concrete existence proof of that hybrid: the round-14 scaffold remediation — output-schema coercion plus fail-loud decision retries, a thin governance layer added to the autonomous path — eliminated both models’ hold-

collapses outright while preserving autonomy’s composite advantage (§7.11), exactly the direction this analysis recommends. Whether an LLM is best deployed as a *disciplined expert-human proxy* (structured) or trusted to *exceed human heuristics* (autonomous) is therefore not a single answer but a function of the model’s self-governance capability — which the scorecard measures directly. We pose this as an open, evidence-informed question rather than a prescription; settling it is a natural use of the instrument as the model panel and the live horizon grow.

8.4 Test-Time-Compute and Benchmark Hygiene

The output-budget finding (§7.8) is a methodological warning for the field: a reasoning model under a tight output cap can look like an incapable portfolio manager when it is merely a *truncated* one. The fix is operationally simple (budget generously; billing is on generated tokens) and the telemetry is mandatory (cap-hits, truncation, parse-fail). A benchmark that does not separate these will misattribute a budget artifact to model capability — the exact error the truncation / rate-limit telemetry corrects (qwen3’s incompleteness, §7.5).

8.5 Cost Realism and Auditability

The cost-tier ablation surfaced a measurement defect: the HIGH (Almgren–Chriss) tier charges an implausible $\sim +527$ bps/round, so it is mis-calibrated rather than “pessimistic.” We retain LOW — its commission and spread legs validated against live broker paper fills (§5.2), its impact leg parameterized rather than fitted — and leave HIGH-tier recalibration to future work. The recompute-verifiable chain (§4.4) provides an integrity check over the persisted store: verification re-derives the hashes and checks every prior-round link. As scoped in §4.4, it does not provide externally anchored non-repudiation.

8.6 Behavioral Interpretability from the Audit Trail

Because every decision artifact is persisted and hash-chained, CLQT supports a qualitative, reproducible read of how models reason and behave (method §6.5; results §7.11) — intent-vs-clip gaps, Risk-Officer negotiations, tool-foraging order, and lesson formation/reuse, each anecdote resolving to a hash-verified round. We frame this as behavior plus self-report, not hidden chain-of-thought, complementary to the axes: the axes quantify *where* a configuration fails; the behavioral read explains, in the model’s own words and actions, *how*. We regard it as a primary use of the audit trail, not a side effect.

8.7 Limitations

We read this single-period campaign through the standard threats-to-validity frame: what the design can identify, what its constructs assume, and the conditions under which its conclusions hold.

- **What returns can identify.** By construction the returns axis is underpowered to attribute module value at this scale: every single-module *and* cluster knockout lands inside the ± 0.42 repeated-run band (§7.7), so no module-importance claim can rest

on returns — we read the flat ΔSharpe as a statement about the metric, not the modules, and confine module value to the capability axes (where, e.g., analysis-blind drives Acuity to ≈ -0.01). Two related scope conditions: the HIGH cost tier is mis-calibrated (~ 527 bps/round; LOW retained), and Acuity and Discipline are mode-confounded, so they are read only within mode.

- **Construct validity of the axes.** The five axes operationalize latent capabilities, and the v2 hardening of Discipline and Composure shows the read is sensitive to how those constructs are defined. We argue construct validity empirically rather than assume it — rank-stability across the D3/D4 revision ($\tau = 0.86$, $\rho = 0.94$ on the current cohort; $\tau = 0.78$, $\rho = 0.90$ as originally measured) and across the v1.1 Acuity noise-set widening (composite $\tau = 0.75$, $\rho = 0.88$; Appendix F.1b), near-orthogonality (mean $|r| = 0.23$), and robustness to leave-one-out and judge-reweighting (Appendix F) — and note candidly that the earlier revision overturned one prior sub-finding (memory “breaking” Composure was an artifact of the earlier construction).
- **Judge independence.** A held-out, non-cohort judge mitigates but cannot fully eliminate LLM-as-a-judge self-preference. A second, different-family judge corroborates the headline gap on the backtest (GLM-5.2, $\rho = 0.65$ over 633 shared rounds; gap $+0.27$ vs minimax-m3’s $+0.30$). The v1.0 residual — minimax-m3 also sits in the live cohort and self-adjudicated its own live rounds — is resolved in v1.1: each track now has its own out-of-cohort primary (backtest: minimax-m3; live: GLM-5.2), and the scorecard tooling refuses a judge that matches any scored run. What remains is the two-judge panel itself; a broader judge panel is future work (§8.8, Appendix C Table C1).
- **External validity: one regime, modeled backtest costs.** The window is a single, predominantly rising market (SPY $+27\%$, max DD -8.9% , VIX median 17), and backtest costs are modeled rather than executed (the live track supplies real fills on the daily deltas). This bounds every returns-based statement to one regime, and the bound cuts both ways: the benign window *masks* the modules whose value concentrates at turning points and *flatters* high-exposure risk-taking while penalizing the prudent de-risking the agents exhibit (§7.7). The capability and risk-adjusted axes are comparatively regime-robust, but absolute-return conclusions should be read as regime-conditional.
- **Live evidence: consistency, not transfer.** The live track shows the instrument measures the same decision property on genuinely unseen data, not that any specific model generalizes — the live and backtest cohorts are different model versions, and live returns are indicative-only by design (daily flatten; overnight exposure is virtual and modeled). The four-week horizon keeps Acuity and Composure the shortest-baseline axes; tellingly, each extension of the live horizon revised a short-horizon read — week 2 converted an apparent day-2 hold “collapse” into an intermittent, recoverable rate, weeks 3–4 swung gemini’s Composure from the top back to the floor, and the final week lifted deepseek’s and gpt’s autonomous Composure (§7.10–§7.11) — a standing caution against over-reading any short live record, ours included. Two further disclosures bound the live read (§7.10): a mid-campaign prompt/failure-handling remediation splits the campaign at round 14 — pooled results are primary, and the one visibly boundary-sensitive metric

(the autonomous hold rate) is reported split, where the boundary doubles as an intervention test (§7.11) — and one model’s final round ran at reduced reasoning effort after a non-terminating reasoning loop, so the live panel is not perfectly homogeneous in prompt or effort conditions.

None of these undercut the contribution — the instrument and its read-out — but all bound what the *numbers* may claim.

8.8 Future Work

The instrument is built to be populated over time; the priorities are the dimensions along which this paper’s reading is thinnest, not a longer list of run parameters.

- **Scale and statistical power.** Broaden the model panel and the number of repeated runs so every axis carries confidence intervals and rank-stability rather than point estimates and run-ranges — turning the current map into one with quantified uncertainty.
- **Regime robustness.** This campaign sat in a single benign, rising market. Stratifying rounds by market regime and reporting how far each capability degrades from calm into stress is the load-bearing extension before any returns-based conclusion can generalize.
- **Generalization of the profile.** Varying the standing mandate and the reasoning-effort budget tests whether a model’s capability fingerprint is a stable property or an artifact of the single mandate and effort level used here.
- **Active stress and instrument hardening.** Adversarial misleading-data and canary probes would stress Acuity and Composure directly rather than relying on naturally-occurring noise; a broader held-out judge panel, a recalibrated HIGH cost tier, and a judge drawn from outside the live cohort would further harden the measurement itself.

Finer-grained per-sector and per-name attribution is a natural next layer, letting the same audit trail localize not only *where* a process fails but on *which* exposures.

8.9 Ethics and Broader Impact

CLQT evaluates agents on historical data and broker *paper* trading only; no real capital is deployed, and nothing in this paper claims any evaluated configuration is fit for deployment — the findings argue the opposite. The dual-use question for a trading-agent benchmark is whether it functions as a recipe for deploying LLM traders; our results cut against that reading: a cohort-wide coherence failure under an independent judge, model-specific reliability deficits invisible to returns, and scaffold-dependent decision collapses are precisely the evidence a responsible pre-deployment gate should demand, and we frame the instrument as that gate. At the market level, correlated decisions from similar foundation models raise legitimate herding and stability concerns; an instrument that measures overreaction (Composure) and mandate adherence (Discipline) is a tool for *studying* that risk rather than amplifying it. The recompute-verifiable audit trail (§4.4) supplies a decision-level integrity record that can support regulatory review of AI

in finance. All market, news, and macro data are obtained from commercial and public APIs under their terms of service; no personal data is collected or processed.

9. Conclusion

CLQT is a closed-loop, cost-aware, temporally gated benchmark for diagnosis before ranking. Its recompute-verifiable audit trail and five-axis scorecard characterize competencies, limitations, and unexpected behavior in autonomous portfolio management. In the year-long campaign, the capability-composite leader differs from the nominal Sharpe winner, whose 9.82 Sharpe rests on only 5 of 26 rounds, and no configuration dominates all five axes. The audit trail also exposes reliability and stability failures that returns conceal. In the completed four-week live track, the signal–action–agreement and judge components retain a similar descriptive divergence (+0.30 backtest / +0.23 live; protocol-clean within-judge band +0.23...+0.30). This is instrument-consistency across different cohorts, not same-model transfer or a calibrated effect size. The live trail further separates two autonomous hold mechanisms, and their disappearance after the bundled mid-campaign remediation provides targeted, mechanism-consistent intervention evidence. A credible ranking still requires more models, regimes, and repeated runs. CLQT supplies the diagnostic profile needed to make such rankings scientifically interpretable.

References

- [1] Chen, Y., et al. (2025). StockBench: Can LLM Agents Trade Stocks Profitably In Real-world Markets? arXiv:2510.02209.
- [2] Yu, H., Li, F., & You, J. (2025). LiveTradeBench: Seeking Real-World Alpha with Large Language Models. arXiv:2511.03628.
- [3] Fan, T., et al. (2025). AI-Trader: Benchmarking Autonomous Agents in Real-Time Financial Markets. arXiv:2512.10971.
- [4] Li, C., Shi, Y., Luo, Y., & Tang, N. (2025). Will LLMs be Professional at Fund Investment? DeepFund: A Live Arena Perspective. arXiv:2503.18313.
- [5] Li, H., et al. (2024). InvestorBench: A Benchmark for Financial Decision-Making Tasks with LLM-based Agent. ACL 2025. arXiv:2412.18174.
- [6] Li, W. W., Kim, H., Cucuringu, M., & Ma, T. (2025). Can LLM-based Financial Investing Strategies Outperform the Market in Long Run? (FINSABER). arXiv:2505.07078.
- [7] Zhao, Y., Chen, S., & Su, N. (2026). PortBench: A Correlation-Aware, Full-Pipeline Benchmark for LLM-Driven Portfolio Management. arXiv:2605.27887.
- [8] Li, X., et al. (2025). Profit Mirage: Revisiting Information Leakage in LLM-based Financial Agents. arXiv:2510.07920.
- [9] Zhu, T., et al. (2026). From Knowing to Doing: A Memory-Controlled Benchmark for LLM Trading Agents on Stock Markets. arXiv:2605.28359.

- [10] Xiao, Y., Sun, E., Luo, D., & Wang, W. (2024). TradingAgents: Multi-Agents LLM Financial Trading Framework. arXiv:2412.20138.
- [11] Yu, Y., et al. (2023). FinMem: A Performance-Enhanced LLM Trading Agent with Layered Memory and Character Design. arXiv:2311.13743.
- [12] Cao, H., Driouich, I., & Thomas, E. (2026). Beyond Task Completion: Revealing Corrupt Success in LLM Agents through Procedure-Aware Evaluation. arXiv:2603.03116.
- [13] Zheng, L., et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *NeurIPS Datasets & Benchmarks*. arXiv:2306.05685.
- [14] Liang, P., et al. (2022). Holistic Evaluation of Language Models (HELM). *TMLR* (2023). arXiv:2211.09110.
- [15] Yao, S., et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. *ICLR*. arXiv:2210.03629.
- [16] Shinn, N., et al. (2023). Reflexion: Language Agents with Verbal Reinforcement Learning. *NeurIPS*. arXiv:2303.11366.
- [17] Park, J. S., et al. (2023). Generative Agents: Interactive Simulacra of Human Behavior. *UIST*. arXiv:2304.03442.
- [18] Packer, C., et al. (2023). MemGPT: Towards LLMs as Operating Systems. arXiv:2310.08560.
- [19] Sumers, T., Yao, S., Narasimhan, K., & Griffiths, T. (2024). Cognitive Architectures for Language Agents (CoALA). *TMLR*. arXiv:2309.02427.
- [20] Anthropic (2024). Model Context Protocol (MCP): An Open Standard for Connecting AI Assistants to Tools and Data. <https://modelcontextprotocol.io>
- [21] Qin, Y., et al. (2024). ToolLLM: Facilitating LLMs to Master 16000+ Real-World APIs. *ICLR*. arXiv:2307.16789.
- [22] Hong, S., et al. (2024). MetaGPT: Meta Programming for a Multi-Agent Collaborative Framework. *ICLR*. arXiv:2308.00352.
- [23] Liu, X., et al. (2024). AgentBench: Evaluating LLMs as Agents. *ICLR*. arXiv:2308.03688.
- [24] Yao, S., et al. (2024). τ -bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. arXiv:2406.12045.
- [25] Anderson, J. R. (1983). *The Architecture of Cognition*. Harvard University Press. (*ACT**; *declarative vs. procedural knowledge, compilation through practice.*)
- [26] Boyd, J. R. (1987). *A Discourse on Winning and Losing*. Air University. (*The OODA decision cycle.*)
- [27] Almgren, R., & Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5–39.

Appendix A: Cost Tier Parameterizations

Tier	Spread (bps)	Commission (bps)	Slippage (bps)	Borrow (ann bps)	Impact Model
zero	0	0	0	0	none
low (<i>default</i>)	2.0	0.5	1.0	30	square_root
medium	3.0	1.0	2.0	50	square_root
high	5.0	2.0	5.0	100	almgren_chriss

Table A1. *Cost tier parameterizations. Live broker paper fills (§5.2) confirm the LOW-tier flat commission of ~\$1 USD $\approx$ 0.5 bps on a \$200k+ position. Caveat: the HIGH (Almgren–Chriss) tier as parameterized produces ~+527 bps/round in the campaign (§7.7) and must be recalibrated before use.*

Appendix B: SCOUT Signal Computation and MCP Tool Inventory

Momentum ($m_{12,1}$): $(P_{t-21}/P_{t-252}) - 1$ on adjusted closes (1-month skip-lag avoids short-term reversal). Earnings surprise (ϵ_{eps}): $(\text{EPS}_{\text{act}} - \text{EPS}_{\text{est}})/|\text{EPS}_{\text{est}}|$, clamped to $[-5, +5]$; None/excluded when no estimate. Macro correlation (ρ^{mkt}): 60-day rolling Pearson of daily log returns vs SPY, with the <10-obs / near-zero-variance guard described in §3.3.

All quantitative analysis (§3.1) is performed through 19 tools defined to the MCP standard and grouped into four namespaces. The benchmark invokes them *in-process* against the underlying quant/data libraries — we use the MCP tool-definition interface and schemas, not a networked server transport — and every call is hashed, timed, and logged, so tool *use* is itself auditable.

Namespace	Tools
<i>clqt-market-data</i>	<i>fetch_market_bars, fetch_market_snapshot, get_universe, fetch_fundamentals</i>
<i>clqt-quant-tools</i>	<i>compute_signals, compute_risk_report, optimize_portfolio, estimate_transaction_costs, compute_portfolio_analytics, compute_covariance_matrix, compute_factor_attribution, compute_regime_performance, compute_alpha_beta</i>
<i>clqt-news-sentiment</i>	<i>fetch_news, fetch_sentiment, fetch_macro_indicators</i>
<i>clqt-execution</i>	<i>simulate_execution, execute_orders, get_execution_report</i>

Table B1. *Tool inventory — 19 tools defined to the MCP standard, grouped into four namespaces and invoked in-process.*

Appendix C: Model Registry — IDs, Snapshots, and Training Cutoffs

Model (OpenRouter id)	Provider	Knowledge cutoff	Group
deepseek/deepseek-chat-v3.1	DeepSeek	Mar 2025	Backtest
google/gemini-2.5-flash-lite	Google	Jan 2025	Backtest
openai/gpt-5-mini	OpenAI	May 2024	Backtest
anthropic/claude-haiku-4.5	Anthropic	Feb 2025 ‡	Backtest
qwen/qwen3-235b-a22b-2507	Alibaba	Jun 2025	Backtest
deepseek/deepseek-v4-pro	DeepSeek	not disclosed †	Live
google/gemini-3.5-flash	Google	Jan 2025	Live
openai/gpt-5.4-mini	OpenAI	Aug 2025	Live
anthropic/claude-sonnet-4.6	Anthropic	Aug 2025 ‡	Live

Model (OpenRouter id)	Provider	Knowledge cutoff	Group
qwen/qwen3.7-plus	Alibaba	not disclosed †	Live
minimax/minimax-m3	MiniMax	not disclosed †	Live · Judge
z-ai/glm-5.2	Z.ai	late 2025 §	Judge

Table C1. *Model registry for the backtest cohort, the live cohort (§7.10), and the held-out Coherence judges (§6.4.1), all accessed via **OpenRouter** under the **provider/model** ids shown (dated suffixes pin the snapshot). Reasoning: backtest per §7.2 (gpt-5-mini effort low, the rest non-thinking/off); live at medium (§7.10; disclosed deviation — qwen3.7-plus’s final round ran with reasoning off after a non-terminating reasoning loop at medium). Cutoffs are provider-stated where published; ‡ = Anthropic’s reliable cutoff (Claude training extends to ~Jan 2026, still pre-live); § = aggregator-reported approximate; † = no cutoff disclosed — but the contamination argument needs no tabulated date: training precedes each model’s 2026 release, all well before the 2026-06-15 live start, with integrity enforced by executing on post-cutoff data (cutoff gate + exchange-calendar exclusion). Backtest cutoffs precede the 2025-06-16 start (qwen3-235b’s Jun 2025 within the ≤ 1 -month grace, §7.2). Judge assignment (v1.1): minimax-m3 is the primary judge for the backtest cohort only (it is not in it); the live panel — which contains minimax-m3 — is judged by GLM-5.2, non-cohort on both tracks. The assignment is enforced by the scorecard tooling, which refuses a judge matching any scored run. (In v1.0, minimax self-judged its own live rounds and scored itself among the lowest, ≈ 0.13 – 0.25 — no self-preference inflation was observed — but the enforced hold-out removes the concern by construction, §8.7.)*

Appendix D: Audit-Trail Field Map for the Scorecard

Every metric in this paper is recoverable from the persisted per-run store without new instrumentation.

Source field(s)	Axis / metric powered
<code>portfolio_after.nav/weights/leverage/cash</code>	returns, vol, Sharpe/Sortino/Calmar, MaxDD
<code>execution_report.{turnover,total_cost,commission,slippage,market_impact,spread}</code> <code>mapc_delta_proposed + signals</code>	cost decomposition; Composure (turnover) Coherence (agreement), Acuity (informative–noise Spearman)
<code>research_brief / target_allocation.rationale + target_weights</code> <code>consistency_score.{composite_score,style_drift} + realized_vol</code> signal	Coherence (judge prompt) Composure (drift, overreaction slope)
<code>target_allocation.constraints_satisfied</code> (pre-MAPC), <code>mapc_adjustments_clipped</code> , turnover vs cap <code>parse_failures</code> , <code>decision_attempts</code> , <code>agent_costs[]</code> .out_tokens (cap-hits), round count	Discipline
<code>agent_costs[]</code> (in/out/cache/reasoning tokens, elapsed_s, cost_usd)	Reliability + truncation telemetry
<code>observation_hash/action_hash/prev_round_hash/signed + IntegrityChecker</code> <code>benchmark_metrics.benchmark_comparisons[]</code> <code>research_brief</code> , <code>analyst_brief</code> , <code>target_allocation.rationale</code> , <code>risk_officer_notes</code> , <code>reflection_notes</code> <code>lessons_generated[]</code> (observation/implication/type/event_types/ regime), <code>memory_provenance</code> , <code>times_referenced</code> <code>mapc_delta_proposed</code> vs <code>mapc_delta_projected</code> , <code>mapc_adjustments_clipped</code> <code>risk_approved</code> , <code>accepted_rejection</code> , <code>no_trade_accepted</code>	efficiency; Behavioral (effort allocation per role) integrity (§4.4)
	benchmark-relative, “beat /8” Behavioral (stated-reasoning corpus, §6.5/§7.11) Behavioral (metacognition / learning trajectories) Behavioral (intent-vs-guardrails); Discipline Behavioral (inter-agent / committee dynamics)

Source field(s)	Axis / metric powered
<code>tool_calls_log[]</code> (tool, order, turns), <code>candidate_signals</code> , <code>universe_changes</code>	Behavioral (information-foraging fingerprint)
<code>decision_attempts</code> , <code>parse_failures</code> , <code>constraint_violations</code>	Behavioral (struggle signatures); Reliability

Table D1. *Audit field $\rightarrow$ metric map. Tooling: `build_paper_tables.py` (body tables, D1–D5 scorecard, and held-out judge outputs), `run_ablation.py` (variant generation + launch), `show_audit.py` –verify (chain verification).*

Appendix E: Behavioral Anecdotes (round-id-pinned)

Representative behaviors from the decision trail, each resolvable to a hash-verified round (the full bank is available from the authors on request). Read neutrally — the same audit field that exposes a failure mode credits a competence.

Module-detach collapses the decision process (the behavioral diagnostic in §7.7). On one round (2025-11-03), same anchor model: the *full* agent weighs ~128 signals, scouts ~35 candidates, and commits a mean-variance-optimal book with explicit weights, an exclusion list, binding constraints, and a cost-management lesson; the *bare-workflow* agent weighs ~56 signals, scouts none, and its entire rationale is a one-line note accepting a default risk-parity book with no constraints, no adjustments, and an empty reflection — premature construction, presumptuous intent, absent scrutiny, yet the caps keep the realized book safe, so returns are unaffected (Table 8).

Two roads to a hold (autonomous mode). *gemini-3.5-flash* (2026-06-16) holds a full mean-variance thesis in its reflection, but its committed round records only a truncation marker — it exhausted its tool-turn budget exploring and never serialized a final allocation. *minimax-m3* (2026-06-16) emits schema-invalid output (an empty value where a list was required); the round is rejected and falls back to hold.

Competent, signal-grounded decisions. *claude-sonnet-4.6* (2026-06-15) reads a risk-on regime, flags a negative momentum loading in its factor model, and rotates to correct it, exiting its worst-momentum and highest-volatility names. *deepseek-v4-pro* (2026-06-17) rotates from deteriorating mega-cap tech into cyclicals and financials, citing a large-cap sentiment collapse against improving trajectories elsewhere. *gpt-5.4-mini* (2026-06-17) tilts to the strongest-momentum names, keeps an overweight where the news trend improved, and trims negative-momentum or deteriorating-sentiment ones.

These illustrate the two hold mechanisms (§7.11) alongside the signal-grounded reasoning the scorecard rewards (*claude-sonnet-4.6* Coherence 0.60, APM-CS 83.6).

Appendix F: Capability-Axis Robustness (construct validity)

This version hardens two axes (§6.4) and validates that the change does not rewrite the conclusions. Discipline (D4) is now bounded, $\frac{1}{2}\text{self_constrained} + \frac{1}{2}(1 - \text{clip_dependence})$ — the earlier draft scored only the pre-MAPC self-constraint rate plus a turnover-cap leg that was a no-op at campaign turnover (~0.15). Composure (D3) is now the mean of its legs rather than their product, which let one weak leg zero the axis and (since the

consistency leg comes from the memory module) made Composure *uncomputable* under the memory ablation — the mean falls back to the memory-free leg. All numbers come from *scripts/analyze_axis_robustness.py* over the per-run scorecard.

F.1 — Rank stability (v1 $\rightarrow$ v2, layered). On the current five-seed judged scorecard, isolating each revision and comparing config-mean APM-CS orderings: the D3/D4 revision alone gives Kendall $\tau = 0.86$ / Spearman $\rho = 0.94$ (its original v0.8.1 measurement on the three-seed cohort was $\tau = 0.78$ / $\rho = 0.90$, $p < 0.001$); the v1.1 Acuity noise-set revision alone gives $\tau = 0.75$ / $\rho = 0.88$ (F.1b); compounding all three axis revisions at once gives $\tau = 0.58$ / $\rho = 0.78$. The main claims are unchanged under every variant — deepseek · structured leads APM-CS and qwen3 · structured is the Reliability artifact under *both* constructions. The movement is where the fixes aim: configurations v1 Discipline over-credited (a perfect self-constraint rate without the clip-dependence penalty), and D2 re-ranks the mid-pack once attention to the horizon-mismatched RSI distractor is penalized. The D3/D4 revision overturned exactly one prior sub-finding — that removing memory “breaks” Composure — an artifact of the old uncomputable construction; under v2, memory removal merely drops the consistency leg (§7.7).

F.1b — Acuity noise-set revision (v1.1). Widening the D2 noise leg from realized-vol alone to the two-distractor composite mean($z(\text{realized_vol})$, $z(\text{RSI}_{14})$) (§6.4) re-orders the D2 axis moderately (config-mean D2 ordering: Kendall $\tau = 0.56$, Spearman $\rho = 0.78$ across the ten configurations) but leaves the APM-CS composite ordering essentially intact ($\tau = 0.75$, $\rho = 0.88$ on the judged five-seed scorecard; D1 held fixed between variants, so the movement isolates the noise-set change). The single substantive re-read is haiku · structured, whose Acuity rises from ≈ 0.2 to 0.44 — its proposals avoid the RSI distractor that the single-leg test did not probe (measured on the single ≥ 3 -tilt round its sparse-proposal style yields, §7.3-F4). The single-leg variant is exported as *D2_acuity_v1* alongside the headline axis, mirroring the D3/D4 v1 $\leftrightarrow$ v2 treatment.

F.2 — Orthogonality (non-redundancy). Across the ten off-diagonal axis pairs the mean |Pearson r | is 0.23 (max 0.55, Coherence $\leftrightarrow$ Discipline; Figure F1), so the five axes are near-orthogonal — they measure distinct properties rather than restating one underlying score. The strongest pair (Coherence vs. Discipline, $r \approx -0.55$) is itself interpretable: configurations that self-constrain hardest are not the ones whose allocations best follow their stated reasoning.

	Coher.	Acuit.	Compo.	Disci.	Relia.
Coherence	1.00				
Acuity	0.22	1.00			
Composure	-0.24	0.06	1.00		
Discipline	-0.55	0.25	0.34	1.00	
Reliability	0.11	0.37	-0.10	-0.10	1.00

Figure F1. *Inter-axis Pearson correlation across the campaign cohort (D1–D5). Off-diagonal magnitudes are small (mean $|r| = 0.23$), evidence the five axes are non-redundant.*

F.3 — Weight sensitivity. Re-aggregating APM-CS under leave-one-axis-out (five variants) and under reweighting the two Coherence components (agreement vs. judge from 0.3/0.7 to 0.7/0.3) leaves the configuration ordering stable: Kendall $\tau \geq 0.71$ against the equal-weight baseline in every case (the Coherence-reweighting variants stay at $\tau \geq 0.89$). The headline ordering is therefore a property of the cohort, not of the particular axis weights.

F.4 — Cross-mode sensitivity on the mode-comparable axes. Because Acuity (D2) and Discipline (D4) are mode-confounded (§6.4), the cross-mode APM-CS contrasts in the body are descriptive profile summaries. As a sensitivity, Table F1 restricts the comparison to the three axes measured identically in both modes — Coherence (D1), Composure (D3), and Reliability (D5) — averaging their raw values (all bounded to $[0,1]$; equal weights; computed directly from the published per-axis values in Tables 5 and 10). This is a robustness read, not an official composite. On it, the backtest picture is model-dependent rather than uniformly pro-structure: structured is favored only for deepseek (+0.14), haiku and gpt-5-mini are within ± 0.02 , and gemini and qwen3 favor autonomy (qwen3’s structured leg is the five-round showcase seed). The live cohort favors autonomy for all six pairs (margins +0.02 to +0.20). The contrast with the full composite — which favors the four reliable structured backtest configurations — indicates that the composite’s structured advantage on the backtest rides substantially on the mode-confounded axes, consistent with the scoped reading in §7.3-F2 and §7.6.

Table F1. *Cross-mode sensitivity on the mode-comparable axes: per-configuration means of the raw D1/D3/D5 values (from Tables 5 and 10) and their three-axis average; $\Delta = \text{structured} - \text{autonomous}$. ★ qwen3-235b · structured is the five-round showcase seed (§7.3).*

Cohort · model	D1 s/a	D3 s/a	D5 s/a	mean s/a	Δ (s–a)
BT · deepseek-v3.1	0.45 / 0.42	0.93 / 0.90	0.99 / 0.62	0.79 / 0.65	+0.14
BT · claude-haiku-4.5	0.23 / 0.44	0.95 / 0.81	1.00 / 0.98	0.73 / 0.74	−0.02
BT · gpt-5-mini	0.43 / 0.47	0.92 / 0.92	1.00 / 0.99	0.78 / 0.79	−0.01
BT · gemini-2.5-flash-lite	0.38 / 0.47	0.93 / 0.77	0.61 / 0.95	0.64 / 0.73	−0.09

Cohort · model	D1 s/a	D3 s/a	D5 s/a	mean s/a	Δ (s-a)
BT · qwen3-235b ★	0.33 / 0.42	0.95 / 0.91	0.19 / 0.99	0.49 / 0.77	−0.28
Live · claude-sonnet-4.6	0.52 / 0.60	0.93 / 0.94	0.82 / 1.00	0.76 / 0.85	−0.09
Live · deepseek-v4-pro	0.50 / 0.62	0.95 / 0.95	1.00 / 1.00	0.82 / 0.86	−0.04
Live · qwen3.7-plus	0.41 / 0.59	0.95 / 0.94	0.58 / 1.00	0.65 / 0.84	−0.20
Live · gpt-5.4-mini	0.48 / 0.56	0.49 / 0.69	0.97 / 0.99	0.65 / 0.75	−0.10
Live · gemini-3.5-flash	0.42 / 0.51	0.46 / 0.44	0.94 / 1.00	0.61 / 0.65	−0.04
Live · minimax-m3	0.41 / 0.56	0.47 / 0.45	0.56 / 0.49	0.48 / 0.50	−0.02

Appendix G: Reproducibility, Artifacts, and AI-Involvement Statement

Reproducibility. Every number in this paper recomputes from the persisted per-run audit stores: the scorecard builder derives the campaign and live scorecards (Tables 5 and 10) and the D1–D5 axes from the DecisionRound records (field map in Appendix D), the ablation tooling derives the ablation tables (Tables 6–8) from the same stores, the behavioral analyzer derives the hold-mechanism split and the anecdote bank, and the figure scripts regenerate all data figures from those outputs — no reported number is hand-maintained (Table 9’s per-module verdicts synthesize that evidence). Judge scores are cached by *round_id* per judge model (minimax-m3 for backtest, GLM-5.2 for live), so recomputation reuses the exact adjudications reported here rather than re-sampling them. Campaign configuration is declarative (TOML): the backtest ran 26 bi-weekly rounds per seed (five seeds per structurally-reliable configuration, §7.2) and the live track ran the four-week daily campaign of §7.10; model identities and snapshots are pinned in Table C1 (all accessed via OpenRouter), reasoning budgets and output caps are stated in §7.2/§7.10, and per-round LLM token and dollar costs persist in the audit trail (*agent_costs*). Data providers are named in Table 2; all access is point-in-time-gated (§4.1) with the gathered snapshots persisted per round. Hash-chain verification (§4.4) re-derives every round’s hashes, so a third party holding the stores can confirm the recorded decisions are the ones the engine produced. The audit databases, scorecard and analysis tooling, and figure-generation scripts are available from the authors on request.

AI-involvement disclosure. The LLM agents evaluated in this paper are its objects of study. Separately, LLM coding assistants were used, under continuous human direction and review, for parts of the engineering and manuscript-preparation workflow; all experimental results were produced by the benchmark tooling from the persisted audit stores, all citations were resolved and content-checked against their sources, and the authors verified and take responsibility for all claims.